

PURPOSIVE COMMUNICATION

Communication Process Components		
Component	Description	Example
Sender	Originator of the message or idea.	A teacher explaining a topic in class.
Encoding	Process of converting thoughts into communicable symbols, words, gestures, etc.	Speaking, writing, gesturing.
Message	The content or idea conveyed by the sender.	The topic or instructions being discussed.
Channel	Medium or means through which the message is transmitted.	Face-to-face, phone call, email, social media.
Interference	Any noise or barrier that distorts or blocks the message.	Physical noise, language differences, poor internet, emotional bias.
Decoding	Interpretation of the message by the receiver into meaningful information.	A student understanding the teacher's instructions.
Receiver	The individual or group for whom the message is intended.	Students listening in class, audience in a speech.
Feedback	The receiver's response or reaction to the sender's message.	Asking questions, nodding, replying to an email.
Environment	Physical and psychological setting where communication occurs.	Classroom setup, office environment, emotional atmosphere.
Context	Shared background, situation, or relationship influencing communication.	Cultural context, formal/informal setting, prior experiences between sender and receiver.

Basic Language and Communication Concepts	
Term	Definition
Language	A systematic spoken or written way to communicate
Language Acquisition	Learning a native language naturally without formal study
Language Learning	Studying a new language through formal methods
Language Contact	When speakers of different languages interact and influence each other
Communication	From Latin communicare meaning "to share"; exchanging meaning between people
Nature of Communication	More than words; a dynamic process involving at least two people

Classification of Communication by Context		
Context	Definition	Examples
Intrapersonal	Communication within the self.	Self-reflection, internal monologue, journaling.
Interpersonal	Communication between two or more people.	Dialogue, interviews, counseling.
Extended	Communication using technology tools.	Video calls, webinars, online classes.
Organizational	Communication in formal or professional structures or institutions.	Staff meetings, company memos, reports.
Intercultural	Communication among people from different cultures or background	Diplomacy, global teamwork, multicultural events.
Mass Communication	Communication aimed at a large audience through media.	TV broadcasts, radio, social media posts, newspapers.

Classification of Communication by Mode		
Mode	Definition	Examples
Verbal	Use of words in oral or written form to convey messages.	Conversations, speeches, phone calls, books, emails.
Nonverbal	Use of body language, gestures, facial expressions, and tone to convey meaning.	Nodding, eye contact, posture, hand signals, tone of voice.
Visual	Use of imagery or visual representations to support or replace verbal messages.	Graphs, charts, photos, symbols, maps, slide presentations.

Classification of Communication by Purpose and Style		
Style	Definition	Examples
Formal	Structured and well-planned communication intended to inform, persuade, or entertain a chosen audience.	Business letters, academic writing, public speeches, news broadcasts.
Informal	Casual, unstructured communication typically used in personal or everyday contexts.	Friendly chats, text messages, small talk, casual emails.

Grapevine Communication (Informal Organizational Communication)		
Type	Definition	Example
Single Strand Chain	Message passes from one person to another in a linear manner.	A tells B, B tells C, C tells D.
Gossip Chain	One person tells everyone.	An employee spreads a rumor to all co-workers.
Probability Chain	Random spreading where each person tells others randomly.	Someone hears news and tells whoever they run into.

Cluster Chain	One person tells a few people, who then each tell a few others.	A manager tells 3 supervisors, who each tell their teams.
---------------	---	---

Forms of Nonverbal Communication		
Form	Definition	Examples
Kinesics	Body movement used to communicate.	Gestures, facial expressions, posture, head movements.
Oculesics	Communication through eye behavior.	Eye contact, blinking, pupil dilation, gaze direction.
Haptics	Communication through touch.	Handshake, pat on the back, hugs.
Proxemics	Use of physical space in communication	Personal space, territoriality, seating arrangement.
Chronemics	Use of time in communication	Punctuality, pausing before answering, response delays.
Paralanguage	Vocal elements that accompany speech.	Tone, pitch, volume, speaking rate, sighs.
Appearance	Visual impression that affects perception.	Clothing, grooming, accessories.
Silence	Intentional or unintentional absence of speech to convey meaning.	Awkward pause, respectful silence in a meeting.
Artifacts	Objects or visual features used to convey messages.	Uniforms, badges, logos, religious symbols.

Grapevine Communication (Informal Organizational Communication)		
Type	Definition	Example
Single Strand Chain	Message passes from one person to another in a linear manner.	A tells B, B tells C, C tells D.
Gossip Chain	One person tells everyone.	An employee spreads a rumor to all co-workers.
Probability Chain	Random spreading where each person tells others randomly.	Someone hears news and tells whoever they run into.
Cluster Chain	One person tells a few people, who then each tell a few others.	A manager tells 3 supervisors, who each tell their teams.

Principles of Effective Communication (Kapur, 2020)	
Principle	Description
Trustworthiness	Be credible and honest so others can rely on your words.
Effective Speaking	Deliver messages clearly, confidently, and with proper tone.
Active Listening	Pay attention, show interest, and avoid interrupting.
Good Writing Skills	Write with clarity, accuracy, and appropriate tone.
Good Reading Skills	Understand written content fully and critically.
Objective Judgment	Be fair and rational in analyzing messages.
Value Differences	Respect diverse opinions, cultures, and perspectives.
Avoid Assumptions	Do not judge without facts or clear understanding.
Authenticity	Be genuine in your communication; avoid pretending or manipulating.

Ethical Considerations in Communication	
Ethical Practice	Explanation
Respect Audience	Avoid offensive or harmful language.
Be Considerate of Consequences	Think about the potential effects of your message.
Be Truthful	Avoid lying or spreading misinformation.
Use Information Efficiently	Do not overload or misuse data.
Avoid Falsification	Do not fabricate or alter content dishonestly.
Respect Information Rights	Allow others to access or withhold information as appropriate and lawful.

Linear Models of Communication

Aristotle's Model (300 B.C.)	
Elements	Highlights
One-way process focused on public speaking. Elements: Speaker – source of the message Message – what is being said Audience – target listener Effect – influence on the audience	➤ Considered the first communication model in Western tradition. ➤ Designed for rhetoric and persuasion. ➤ Example: A politician giving a speech to gain public support.

Lasswell's Model (1948)	
Elements	Highlights
Process of communication through five guiding questions: Who – the sender or communicator Says what – the message In which channel – the medium or method used To whom – the audience or receiver With what effect – the result or impact of the message	➤ Used in mass communication and propaganda during WWII. ➤ Focused on media effects. ➤ Example: A TV ad influencing consumer behavior.

Shannon-Weaver Model (1949)	
Elements	Highlights
Known as the Mathematical Model of Communication. Elements: 1. Information Source – origin of the message 2. Transmitter – encodes the message into signals 3. Channel – medium through which the signal travels 4. Noise Source – any distortion or barrier 5. Receiver – decodes the signal 6. Destination – intended target of the message 7. Feedback – later added to reflect two-way communication	➤ Created by engineers to improve telephone systems. ➤ Introduced concept of “noise”. ➤ Example: A phone call with static disrupting message clarity.

Berlo’s SMCR Model (1960)	
Elements	Highlights
Highlights importance of communication skills, attitudes, knowledge, and social systems. SMCR stands for: 1. Source – includes communication skills, attitudes, knowledge, social system, and culture 2. Message – includes content, elements, treatment, structure, and code 3. Channel – senses used in communication: hearing, seeing, touching, smelling, and tasting 4. Receiver – has the same components as the source	➤ Emphasizes that communication effectiveness depends on matching source and receiver backgrounds. ➤ Example: Teacher (source) communicating effectively with students.

Interactional Models of Communication	
Interactional Models of Communication	
Elements	Highlights
Communication is a two-way process where sender and receiver alternate roles. Introduced “Field of Experience” for shared understanding.	➤ First to propose communication as a reciprocal process. ➤ Field of experience = shared background. ➤ Example: Teacher-student Q&A session.

Osgood-Schramm Model (1954)	
Elements	Highlights
Developed by William Scramm, known as the father of mass communication A circular model of communication. Both sender and receiver are simultaneous encoders, decoders, and interpreters.	➤ Emphasizes feedback and continuous exchange. ➤ Used in interpersonal and small-group communication. ➤ Example: Face-to-face dialogue.

Transactional Models of Communication	
White’s Model (1960)	
Elements	Highlights
Involves 8 stages of communication: Thinking, Symbolizing, Expressing, Transmitting, Receiving, Decoding, Feedbacking, Monitoring. Shows communication as a sequential process.	➤ First to use the term “feedbacking”. ➤ Recognizes self-monitoring in communication. ➤ Example: Radio DJ evaluating audience response.

Dance’s Helix Model (1967)	
Elements	Highlights
Communication is continuous, dynamic, and ever-expanding, like a helix. Combines linearity and circularity.	➤ Emphasizes time and development in communication. ➤ Our communication today is influenced by our past. ➤ Example: A friendship growing through continuous interaction.

Barnlund’s Model (1970)	
Elements	Highlights
Multi-layered, real-time communication. Both participants are senders and receivers simultaneously. Elements: 1. P (Person) – each communicator 2. D (Decoding) – interpreting incoming cues 3. E (Encoding) – creating outgoing messages 4. Cpr (Public Cues) – environmental or external signs like gestures, tone 5. Cpv (Private Cues) – internal or personal thoughts, feelings, and intentions 6. Cbeh.nv – Nonverbal behavioral cues (e.g., facial expressions, gestures) 7. Vbeh.v – Verbal behavioral cues (spoken words, tone) 8. M – Message being communicated	➤ Best reflects real-life communication. ➤ Includes nonverbal cues, context, and simultaneous feedback. ➤ Example: Group discussion or team meeting.

Summary Comparison Table		
Model Type	Focus	Example
Linear	One-way message flow	Speeches, TV broadcasts
Interactional	Two-way with delayed feedback	Text messaging, email
Transactional	Two-way with real-time simultaneous interaction	Conversations, team collaborations

Written vs. Spoken Language	
Subcategory	Description
Written Language	Structured, planned, formal, uses complex sentences
Spoken Language	Spontaneous, informal, uses contractions and fillers, includes intonation & pauses

Globalization and Its Impact		
Topic	Explanation	Example
Globalization	Global connection in trade, people, services, culture, and ideas.	Filipino teachers working abroad; K-pop in PH culture.
Effect on Communication	Leads to more intercultural interaction; need for global communication skills.	Online meetings between people from different countries.

Cultural Barriers to Communication		
Barrier	Meaning	Example
Language	People use different languages, slang, or meanings.	“Salvage” in English vs. Filipino usage.
Stereotypes/ Prejudices	Judging others based on assumptions.	Assuming all Asians are good at math.
Behaviors/ Beliefs	Customs or beliefs may differ across cultures.	Not shaking hands for religious reasons.
Norms and Values	What’s right or wrong can vary per culture.	Filipinos value pakikisama; others value assertiveness.
Ethnocentrism	Thinking your culture is better than others.	A boss ignoring local customs when working abroad.
Body Language	Gestures mean different things in different cultures.	Thumbs up = good (US), offensive (Middle East).

Registers of Language	
Subcategory	Description
Frozen Register	Fixed, ceremonial language (e.g., national anthem, religious texts)
Formal Register	Used in official communication, speeches, academic writing
Consultative Register	Professional or polite conversations, teacher-student, doctor-patient
Casual Register	Informal conversation among friends, everyday talk
Intimate Register	Private language between close family or partners, uses personal or pet names

Classification of English Users	
Subcategory	Description
Inner Circle (Native speakers)	Countries where English is the first language (e.g., USA, UK, Australia)
Outer Circle (Second language speakers)	Countries where English is an official or second language (e.g., Philippines, India, Nigeria)
Expanding Circle (Foreign language speakers)	Countries where English is learned as a foreign language (e.g., China, Korea, Japan)

Text Types		
Text Type	Description	Examples
Narrative	Tells a story or recounts events	Short stories, novels, anecdotes
Expository	Explains, informs, or describes	Textbooks, manuals, encyclopedias
Persuasive	Convinces or influences the reader/listener	Advertisements, opinion columns
Descriptive	Creates a picture through detailed observation	Travel blogs, nature articles

Argumentative	Presents a claim with evidence and reasoning	Editorials, debate speeches
Instructive/Procedural	Gives steps or instructions to accomplish something	Recipes, how-to guides

Linguistic Landscape		
Concept	Description	Examples
Linguistic Landscape	Visible language in public spaces	Street signs, billboards, posters
Top-down signs	Official or institutional signs	Government signs, road names
Bottom-up signs	Informal or personal signage	Store signs, graffiti, flyers
Geosemiotics	Study of how placement/ location affects meaning of signs	A "No Parking" sign near driveway
Indexicality	Signs point to context (time, place, people)	A sign in Tagalog near a sari-sari store
Dialogicality	How signs interact or respond to each other	Protest signs replying to billboards
Selection	Choice of language(s), design, and symbols used	Use of English for prestige

Online Landscape		
Element	Description	Examples
Digital Presentation	How information is visually and structurally displayed online	Website layout, app interfaces
Multimodal Elements	Use of text, image, video, audio together	YouTube videos, infographics
Interaction	Audience engagement and feedback	Comments, likes, shares, hyperlinks

Types of Public Signs		
Sign Type	Purpose	Examples
Regulatory Signs	Control or direct behavior	No Smoking, Speed Limit, Keep Out
Infrastructure Signs	Identify structures or services	Hospital, LRT Station, Restroom signs
Commercial Signs	Promote products, services, or businesses	Sale banners, food advertisements
Transgressive Signs	Violate norms, rules, or expectations	Graffiti, protest placards, satire art

Design Tips for Multimedia	
Design Tip	Explanation
Clear visuals	Use relevant images that support the message
Readable text	Choose legible fonts and proper text size
Working links	Ensure hyperlinks and buttons are functional
Purposeful formatting	Use consistent layout, alignment, and colors to guide the viewer’s focus

Pecha Kucha Format and Structure (Presentation)	
Aspect	Details
Definition	A timed presentation format: 20 slides x 20 seconds
Total Duration	6 minutes and 40 seconds
Slide Content	Minimal text, relies heavily on images
Speaking Style	Fast-paced, synchronized with slides

Types of Inquiry Letters	
Type	When Used
Solicited	Responding to an ad or offer
Unsolicited	Asking without being invited
Favor-Seeking	Requesting help or permission

Blogging Basics	
Aspect	Description
Definition	An online journal or article-based platform
Purpose	Share ideas, opinions, updates, or educational content with readers
Platform	Can be hosted on sites like WordPress, Blogger, Medium, etc.

Types of Communication		
Type	Main Purpose	Focus
Informative	To explain or give facts	People, events, things
Persuasive	To convince or motivate	Opinions, actions
Argumentative	To find or prove the truth	Claims with evidence

Public Speaking Types		
Type	What It Is	Used In
Manuscript	Read word-for-word	Speeches, news reading
Memorized	Fully memorized, no script	Contests, performances
Extemporaneous	Prepared with notes	Presentations, talks
Impromptu	No preparation	Recitations, quick talks

Structure of Inquiry Letter	
Part	What to Include
Sender Info	Name, contact, address
Date	When letter was written
Subject	Main reason for writing
Salutation	Greeting (e.g., Dear Sir/Madam)
Body (3 parts)	Purpose → Details → Request
Closing	Polite ending (e.g., Sincerely)

Nature of Communication

1. It is a **continuous process**
2. It involves **two or more people**
3. It is expressed through **verbal and non-verbal means**

Functions of Communication

1. **Control** – regulates behavior and maintains order
2. **Social Interaction** – facilitates relationships and social connections
3. **Motivation** – inspires and encourages action or behavior
4. **Emotional Expression** – conveys feelings and emotions
5. **Information Dissemination** – shares knowledge and data

Globalization And Communication

- **Globalization** – the process of spreading products, ideas, technology, and jobs across countries and cultures
- **Global Village** – a term by **Herbert Marshall McLuhan (1964)** referring to the world becoming interconnected through telecommunications
- **Culture** – shared patterns of behavior, beliefs, values, and norms of a group or society

Characteristics Of Culture

1. **Culture is Learned** – not inherited; acquired through interaction and understanding
2. **Culture is Shared** – involves group behavior and social interactions
3. **Culture is Multi-faceted** – includes cultural universals and norms influencing all aspects of life
4. **Culture is Dynamic** – constantly changing due to time, experience, and external influence
5. **Cultural Identities are Overlapping** – multiple cultural influences may coexist, blend, or sometimes clash

Adaptation To New Cultures

1. **Cultural Diversity** – the presence of multiple distinct cultures within a group or society
2. **Cultural Integration** – cultures exchange traits and influence one another while retaining their uniqueness
3. **Cultural Assimilation** – adopting the dominant culture completely, sometimes losing original identity
4. **Multiculturalism** – coexistence of multiple distinct cultures within a single society
5. **Cultural Accommodation** – adapting to a new culture while preserving one's original culture
6. **Separation** – resisting interaction or integration with another culture; cultural isolation

Kinds Of Responses To Cultural Differences

1. **Avoiding** – refusal to engage or comply with cultural practices that conflict with one's own principles
2. **Accommodating** – accepting the differences in others' ethical systems
3. **Forcing** – imposing one's own ethical views, insisting they are superior
4. **Educating–Persuading** – explaining and promoting one's ethical principles in an effort to influence others
5. **Negotiating–Compromising** – mutual agreement where both parties give up something to resolve a conflict
6. **Collaboration–Problem Solving** – working together to understand and solve issues across cultures

Impact Of Technology On Communication

1. Provides access to vast information and knowledge
2. Facilitates forming new connections and friendships
3. Keeps people connected regardless of distance
4. Introduces cultural diversity into daily life
5. Enhances collaboration between individuals of different cultures
6. Can lead to social isolation and reduced face-to-face interaction
7. May encourage the creation of "second lives" or online identities

Culture Shock

Culture Shock – a feeling of disorientation and discomfort due to sudden immersion in an unfamiliar culture

Four Stages of Culture Shock

1. **Honeymoon Phase** – excitement and fascination with the new culture
 2. **Negotiation Phase** – feelings of frustration and homesickness begin; stereotypes may serve as a coping mechanism
- 3 Main Types of Reaction**
- a. **Rejecters** – isolate themselves and avoid interaction with the new culture
 - b. **Adopters** – fully accept and embrace the new culture
 - c. **Cosmopolitans** – adapt to the new culture while maintaining their original cultural identity
3. **Adjustment Phase** – typically occurs 6 to 12 months after exposure; individuals begin to adapt and feel more comfortable
 4. **Reverse Culture Shock** – difficulty readjusting to one's original culture after adapting to a new one

Local And Global Communication In Multicultural Settings

1. **Multiculturalism** – a society made up of diverse cultural and ethnic groups

- 2. **Intercultural Communication** – effective interaction between people from different cultural backgrounds

Forms Of Intercultural Communication

- 1. **Interracial** – communication between people of different races
- 2. **Interethnic** – communication among people of different ethnic backgrounds
- 3. **International** – communication between representatives from different nations
- 4. **Intracultural** – communication among people of the same racial or ethnic group

Context In Culture

- **High-Context Culture** – communication relies heavily on nonverbal cues, traditions, and implicit understanding
- **Low-Context Culture** – communication is direct, clear, and explicit

Cultural Awareness And Sensitivity

- **Cultural Awareness** – understanding that cultural differences exist
- **Cultural Sensitivity** – recognizing and respecting cultural differences without assuming superiority

Various Cultural And Intercultural Modes

- 1. **Eye Contact** – varies in meaning depending on the culture (e.g., respect, aggression, honesty)
- 2. **Hand Gestures** – gestures can have different meanings across cultures
- 3. **Greetings** – may include handshakes, bows, or verbal expressions

Communication Issues

- 1. **Sexism** – discrimination based on gender
 - **Gender Sensitivity** – awareness and avoidance of gender-based discrimination in communication
- 2. **Racism** – belief in the superiority of one race over another; divides people based on physical characteristics

Political Correctness

Adopting language and behavior aimed at avoiding offense or disadvantage to individuals or groups in society

Language Variety

Language Variety – different forms of a language used by various groups

- 1. **Dialect** – regional or social variation of a language
 - a. **Regional Dialect** – based on location
 - b. **Sociolect** – based on social class or group
 - c. **Ethnolect** – used by a specific ethnic group
 - d. **Idiolect** – individual's unique speech style

- 2. **Accent** – way of pronouncing words
- 3. **Mother Tongue** – first language of a community
- 4. **Pidgin** – simplified mix of languages used to communicate without a shared language

Language Registers

Language Registers – levels or styles of language use

- 1. **Frozen** – fixed, ritualistic (e.g., prayers, oaths)
- 2. **Formal** – structured, professional (e.g., speeches, reports)
- 3. **Consultative** – interaction with expert or authority (e.g., teacher-student)
- 4. **Casual** – everyday conversation among friends
- 5. **Intimate** – private, personal talk (e.g., between close friends or family)

MALAYUNING KOMUNIKASYON SA FILIPINO

Intelektwalisasyon

Barayti ng Filipino na itinuturo sa mga Filipino sa lahat ng antas ng edukasyon mula elementarya hanggang kolehiyo (tulad ng Filipino sa paaralan).

Mga Dimensyon Ng Pagpaplanong Pangwika

- **Seleksyon** – pagpili ng wikang gagamitin (halimbawa, pagpili ng Filipino bilang pambansang wika)
- **Estandisasyon** – pagtatakda ng tuntunin sa wastong gamit (pagbuo ng tuntunin sa balarila at bokabularyo)
- **Propagasyon** – pagpapalaganap ng wikang napili (pagsasama ng Filipino sa kurikulum)
- **Kultibasyon** – pag-aangkop ng wika bilang bahagi ng kultura at lipunan (paggamit ng Filipino sa araw-araw na buhay at kultura)

Teorya Ng Pinagmulan Ng Wika

- **Bow-wow** – wika ay hango sa tunog ng hayop (tunog ng aso, pusa)
- **Ding-dong** – wika ay hango sa tunog ng bagay (tunog ng kampana)
- **Poo-poo** – wika ay hango sa emosyon (sighs, daing)
- **Yo-he-ho** – wika ay hango sa puwersa o pagsusumikap (tunog ng sabayan na paggawa)
- **Mama** – tunog mula sa unang pantig ng "mother" (pagbigkas ng ina ng bata)
- **Sing-song** – wika ay musikal o may melodiya (pag-awit ng mga sinaunang tao)
- **Hey You!** – wika ay ginagamit para sa pagkakakilanlan at pagtawag (pagtawag sa kaibigan)
- **Coo coo** – tunog na gaya ng sa sanggol (tunog ng baby cooing)
- **Yum yum** – tunog na kaugnay sa pagkain (tunog ng gutom)

- **Babble Lucky** – nagkataong mabuo ang mga tunog na nagkaroon ng kahulugan (pagbuo ng mga unang salita)
- **Hocus Pocus** – wika ay mahiwaga o mahikal (ritwal, panalangin)
- **Eurekall** – wika bilang tanda o alaala (pagkilala sa isang bagay)
- **Tata** – wika bilang paalam (pagsabi ng paalam)

Rehistro Ng Wika

- **Lingua Franca** – wikang ginagamit bilang pantulay
 - **Unibersal** – English (halimbawa: komunikasyon sa buong mundo)
 - **Pambansa** – Filipino (wikang pambansa ng Pilipinas)
 - **Rehiyonal** – Ilocano, Cebuano (wikang ginagamit sa mga rehiyon)
- **Dayalek** – barayti ng wika batay sa lugar (halimbawa: Batangas Tagalog vs. Metro Manila Tagalog)
- **Idyolek** – natatanging paraan ng pagsasalita ng isang tao (style ng pagsasalita ng isang indibidwal)
- **Sosyolek** – wika batay sa grupo ng lipunan (wikang ginagamit ng mga kabataan, propesyonal)
- **Ekolek** – wikang ginagamit sa loob ng tahanan (wikang ginagamit ng pamilya)
- **Etnolek** – wika ng partikular na pangkat etniko (halimbawa: Kapampangan, Ifugao)
- **Jargon** – teknikal na salita sa isang propesyon o larangan (medical terms, legal terms)
- **Pidgin** – pinaghalong wika para sa komunikasyon ng magkaibang grupo (halimbawa: Bisaya-English pidgin)
- **Creole** – nativized o unang wika mula sa pidgin (halimbawa: Chavacano)

Antas Ng Wika

- **Pormal** – ginagamit sa opisyal na okasyon o sulatin
 - ✓ **Pambansa** – salitang pang-araw-araw at kilala ng lahat (halimbawa: "upuan")
 - ✓ **Pampanitikan** – mas mataas na antas at karaniwang ginagamit sa panitikan (halimbawa: "silya")
- **Impormal** – pang-araw-araw na gamit
 - ✓ **Balbal** – salitang kalye (halimbawa: "jologs")
 - ✓ **Kolokyal** – karaniwang salita na may halo ng ibang wika (halimbawa: Taglish, Enggalog)
 - ✓ **Lalawiganin** – salita mula sa mga lalawigan (halimbawa: "kadiwa")
 - ✓ **Bulgar** – bastos o di-kanais-nais na pananalita (halimbawa: mura)

Pamantayan Sa Pagtataya Ng Wika

- **Pakikibagay** – kakayahang umangkop sa kausap at sitwasyon (halimbawa: pagsasalita ayon sa kaedad o katayuan)
- **Paglahok sa usapan** – aktibong pakikibahagi sa talakayan (halimbawa: pagbibigay-opinyon)
- **Pamamahala sa pag-uusap** – kakayahang kontrolin at panatilihin ang daloy ng komunikasyon (halimbawa: pag-turnover ng usapan)
- **Pagkapukaw ng damdamin** – pagpaparamdam ng empathy (pag-unawa sa damdamin ng kausap)
- **Bisa** – pagiging epektibo ng pagpapahayag (nakukuha ang nais na reaksiyon)
- **Kaangkupan** – pagiging angkop ng wika sa sitwasyon at kausap (tamang gamit ng salita batay sa konteksto)

Komunikasyon At Globalisasyon

- **Monolingwalismo** – paggamit ng iisang wika lamang (halimbawa: Japan)
- **Bilinggwalismo** – paggamit ng dalawang wika (halimbawa: Filipino at English)
- **Multilinggwalismo** – paggamit ng higit sa dalawang wika (halimbawa: Tagalog, Ilocano, English)
- **Polyglot** – taong marunong at kayang gumamit ng tatlo o higit pang wika (halimbawa: interpreter o language expert)

Kulturang Popular

Modernong kultura na mabilis na naipapasa sa pamamagitan ng media, teknolohiya, at iba't ibang plataporma (halimbawa: K-pop, memes, TikTok trends)

Komunikasyon (Talastasan)

Pagpapalitan o pagbabahagi ng kaisipan, damdamin, o impormasyon sa pagitan ng dalawa o higit pang tao

Elemento Ng Komunikasyon

- **Tagapagpadala (Sender)** – pinagmulan ng mensahe
- **Mensahe** – ideya o damdaming nais iparating
- **Tsanel** – daluyan ng mensahe (halimbawa: pasalita, pasulat)
- **Tagatanggap (Receiver)** – tumatanggap ng mensahe
- **Fidbak (Feedback)** – tugon ng tagatanggap

Modelo Ng Komunikasyon

- **Aristotle's Model**
 - Speaker → Speech → Audience → Effect
(naka-sentro sa tagapagsalita at epekto sa tagapakinig)
- **Berlo's SMCR Model**
Source → Message → Channel → Receiver
(kasama ang salik tulad ng attitude, skills, at

kultura sa pagpapadala at pagtanggap ng mensahe)

- **Schramm's Model**
Encoder-Interpreter-Decoder
(binibigyang-diin ang pagkakatulad ng karanasan ng tagapagsalita at tagapakinig)

Uri Ng Komunikasyon

1. **Berbal** – ginagamitan ng salita (pasalita o pasulat)
2. **Di-berbal** – hindi ginagamitan ng salita
 - a. **Proxemics** – distansya o espasyo (lapit o layo ng kausap)
 - b. **Oculesics** – galaw ng mata o paningin (eye contact o pag-iwas ng tingin)
 - c. **Pictics** – ekspresyon ng mukha (pagngiti, pagkunut-noo)
 - d. **Chronemics** – paggamit ng oras (maagang pagdating o pagka-late)
 - e. **Haptics** – paggamit ng haplos o paghipo (tapik sa balikat)
 - f. **Olfactorics** – paggamit ng pang-amoy (amoy ng pabango, pagkain)
 - g. **Objectics** – gamit na may kahulugan sa komunikasyon (bag, cellphone)
 - h. **Colorics** – gamit ng kulay (itim – pormal, pula – tapang)
 - i. **Iconics** – simbolo o imaheng may kahulugan (logo, signage)
 - j. **Kinesics** – kilos ng katawan (kumpas ng kamay, pagtango)

Antas Ng Komunikasyon

- **Intrapersonal** – pakikipag-usap sa sarili (pagninilay)
- **Interpersonal** – pakikipag-usap sa ibang tao (usap ng magkaibigan)
- **Pampubliko** – komunikasyong para sa nakararami (talumpati)
- **Pang-media** – gamit ang mass media (TV, radyo, social media)

Makrong Kasanayan Sa Pagkomunikasyon

- **Pagbasa** – pagbibigay-kahulugan sa mga simbolo o titik
Proseso ng Pagbasa:
 - a. **Persepsyon** – pagkilala sa simbolo (letra/salita)
 - b. **Komprehensyon** – pag-unawa sa binasa
 - c. **Reaksyon** – opinyon o tugon sa binasa
 - d. **Asimilasyon** – pagsanib ng bagong kaalaman sa dating kaalaman
- **Konotasyon** – di-tuwirang kahulugan, madalas masining o malikhain (puso – damdamin)
- **Denotasyon** – tuwirang kahulugan mula sa diksyunaryo (puso – bahagi ng katawan)

Teorya Ng Pagbasa

- **Bottom-up** – nagsisimula sa teksto; unti-unting pag-unawa (mula salita patungo sa kahulugan)
- **Top-down** – mula sa karanasan ng mambabasa patungo sa pag-unawa ng teksto
- **Interaktib** – pinagsamang top-down at bottom-up
- **Iskima** – paggamit ng dating kaalaman upang maunawaan ang bagong impormasyon

Uri Ng Pagbasa

- **Scanning** – paghahanap ng tiyak na impormasyon (numero, pangalan)
- **Skimming** – mabilisang pagbasa para sa pangkalahatang ideya (buod ng kwento)
- **Pasaklaw (Intensive)** – masinsinang pag-unawa sa nilalaman (analisis ng artikulo)
- **Panlibang** – pagbasa para sa aliw o kasiyahan (komiks, nobela)

Pagsulat

Pagsulat – pagsasatitik ng mga kaisipan, damdamin, at ideya sa malinaw, maayos, at makabuluhang paraan

Layunin Ng Pagsulat

- **Persweysib** – nanghihiikayat (hal. patalastas, sanaysay na nangungumbinsi)
- **Informativ** – nagbibigay ng impormasyon (hal. ulat, pamanahong papel)
- **Malikhain** – pampanitikan o masining (hal. tula, maikling kwento)

Uri Ng Pagsulat

- **Akademik** – pormal, may estruktura at lohika, pang-intelektwal (hal. tesis, replektibong sanaysay)
- **Journalistik** – balita at mga artikulo para sa pahayagan (hal. editoryal, lathalain)
- **Teknikal** – tumutugon sa suliraning kognitibo o saykolohikal (hal. manwal, feasibility study)
- **Malikhain** – imahinatibo, masining (hal. dula, tula, kwento)
- **Referensyal** – batay sa sanggunian, ginagamit sa pananaliksik (hal. bibliograpiya, citation)
- **Propesyonal** – eksklusibo para sa isang larangan o propesyon (hal. legal na dokumento, medikal na ulat)

Mga Dapat Isaalang-alang Sa Pagsulat

- **Kaisahan** – isang pangunahing diwa
- **Kasapatan** – sapat na detalye at paliwanag
- **Kaugnayan** – lohikal at maayos na daloy
- **Kalinawan** – malinaw at madaling maunawaan
- **Diin** – pagbibigay-pansin sa mahalagang punto

Proseso Ng Pagsulat

- **Pre-writing** – paghahanda, brainstorming, outline
- **Drafting** – unang pagsulat ng ideya
- **Revising** – pagrerebisa ng nilalaman, organisasyon, at estilo
- **Polishing** – pagwawasto ng grammar, spelling, at estruktura

Bahagi Ng Liham

- **Pamuhatan** – address ng sumulat
- **Petsa** – araw ng pagsulat
- **Patunguhan** – pangalan at address ng tatanggap
- **Bating panimula** – pagbati (hal. Mahal kong Ginoo,)
- **Katawan** – pangunahing nilalaman
- **Bating pangwakas** – pamamaalam (hal. Lubos na gumagalang,)
- **Lagda** – pangalan ng sumulat

Gamit Ng Mga Bantas

- **Kudlit (')** – inaalís na titik (hal. 'di, 'yan)
- **Kuwit (,)** – paghihiwalay ng mga salita o parirala
- **Tulduk-kuwit (;)** – paghihiwalay ng magkakaugnay na kaisipan
- **Tutuldok-tuldok (...)** – patlang sa iniwang pahayag
- **Tuldok (.)** – katapusan ng pangungusap
- **Gatlang (—)** – pagpuputol o pagsingit ng ideya
- **Pasok/Indensyon** – panimulang espasyo sa simula ng talata
- **Padamdang (!)** – matinding damdamin
- **Pananong (?)** – pagtatanong
- **Gitling (-)** – pag-uugnay ng mga salita (hal. araw-araw)

Pagsasalita

Pagsasalita – pagpapahayag ng kaisipan sa pamamagitan ng wika at tinig

Salik Sa Epektibong Pagsasalita

- **Kaalaman** – sapat na impormasyon sa paksa
- **Kasanayan** – kakayahang magsalita nang maayos
- **Tiwala sa sarili** – kumpiyansa habang nagsasalita
- **Tindig** – maayos na postura at kilos habang nagsasalita

Dell Hymes – Speaking Model

Ginagamit upang masuri ang mga salik sa komunikasyon:

- **S – Setting/Scene** (Lugar at oras ng usapan)
- **P – Participants** (Sino ang kalahok)
- **E – Ends** (Layunin ng komunikasyon)
- **A – Act Sequence** (Daloy o pagkakasunod-sunod ng mensahe)
- **K – Key** (Antas ng pormalidad o emosyonal na tono)

- **I – Instrumentalities** (Daluyan – pasalita, pasulat, teknolohiya)
- **N – Norms** (Mga inaasahang asal o panuntunan sa komunikasyon)
- **G – Genre** (Uri ng komunikasyon – talumpati, kwento, debate)

Pakikinig

Pakikinig – Aktibong proseso ng pagtanggap at pag-unawa ng mensahe gamit ang pandinig at pag-iisip

Uri Ng Pakikinig

- **Pasibo/Marginal** – nakikinig pero walang pokus
- **Atentibo** – may konsentrasyon at pokus
- **Kritikal** – sinusuri ang katotohanan ng mensahe
- **Analitikal** – pinag-aaralan ang nilalaman at lohika
- **Apresyatibo** – pakikinig para sa aliw o kasiyahan (hal: musika, kwento)

Uri Ng Tagapakinig

- **Two-eared listener** – nakatuon nang buo, aktibong tagapakinig
- **Eager beaver** – nakikinig pero hindi nagpoproseso, tango lang nang tango
- **Busy bee** – nakikinig habang may ibang ginagawa
- **Tiger listener** – agresibong tagapakinig, kadalasang nanghihimasok
- **Frowner** – laging nakakunot ang noo, palaging hindi sang-ayon
- **Bewildered** – lito, hindi maintindihan ang pinag-uusapan
- **Sleeper** – hindi talaga nakikinig, maaaring antok o walang gana

Istilo Ng Pakikinig

- **People-oriented** – nakatuon sa tagapagsalita, relasyon ang mahalaga
- **Content-oriented** – mahalaga ang detalye at laman ng mensahe
- **Action-oriented** – interesado sa mga konkretong hakbang at resulta
- **Time-oriented** – mahalaga ang oras, gusto ng maikli at diretsong mensahe

Linggwistika

Linggwistika – Agham ng wika at istruktura nito

Mga Sangay Ng Linggwistika

- **Ponolohiya** – pag-aaral ng mga tunog ng wika
- **Morpolohiya** – pagbuo ng salita mula sa mga morpema
- **Sintaksis** – pagsasaayos ng mga salita sa loob ng pangungusap
- **Semantika** – pag-aaral ng kahulugan ng salita at pangungusap
- **Pragmatiks** – gamit ng wika ayon sa konteksto

- **Ortograpiya** – baybay, bantas, at pagsulat ng mga salita

Binibigkas Na Tunog

- **Energy** – puwersang nagmumula sa бага na ginagamit sa pagbigkas
- **Artikulador** – bahagi ng katawan na humuhubog sa tunog (hal: dila, labi)
- **Resonador** – nagpapalakas o nagmomodipika ng tunog (hal: bibig, ilong)

Kaugnay Na Sangay Ng Linggwistika

- **Pragmatiks** – praktikal na gamit ng wika sa konteksto ng komunikasyon
- **Sosyolinggwistika** – pag-aaral ng relasyon ng wika at lipunan
- **Antropolohiyang Linggwistika** – wika bilang bahagi ng kultura at pamumuhay ng tao
- **Saykolinggwistika** – relasyon ng wika at isip o pag-iisip
- **Pilosopikal na Linggwistika** – relasyon ng wika at kaalaman o pilosopikal na pag-unawa

Diptonggo

Diptonggo – Patinig na sinusundan ng malapatinig (w/y) sa loob ng isang pantig
Halimbawa:

- sabaw
- ayaw
- kalabaw

Klaster

Klaster – Kambal-katinig sa loob ng isang pantig (*hindi pinaghihiwalay na katinig*)
Halimbawa:

- plato
- klase
- tren

Morpolohiya

Maagham na pag-aaral ng morpema o makabuluhang yunit ng salita

- Ilang tunog sa alpabeto: **21**
- **Morpemang Panlapi:**
 - **Unlapi** (nasa unahan): **masikap**
 - **Gitlapi** (nasa gitna): **sinikap**
 - **Hulapi** (nasa dulo): **sikap****an**
 - **Kabilaan** (unahan at hulihan): **pinagsikap****an**
 - **Laguhan** (unahan, gitna, at hulihan): **pinagsumikap****an**

Pagbabagong Morpoponemiko

Pagbabago ng tunog kapag pinagdugtong ang mga morpema

- **Asimilasyon** – pagbabago ng ponema ayon sa kasunod (hal: pan, pam, pang)
(hal: panulat, pambansa, pangarap)
- **Pagpapalit Ponema** – pagpapalit ng isang tunog
(hal: bato → batuhin)

- **Metatesis** – palit-posisyon ng ponema
(hal: lipad → nilipad)
- **Pagkakaltas ng Ponema** – pagtatanggal ng tunog
(hal: taniman → tamnan)
- **Reduplikasyon** – pag-uulit ng bahagi ng salita
(hal: takbo → tatakbo)
- **Paglilipat ng Diin** – pagbago ng kahulugan ayon sa diin
(hal: *BÁsa* [noun] → *basÁHIN* [verb])
- **Pag-uulit ng Salita**
 - **Parsyal** – bahagi lang ang inuulit (minu-minuto, bali-balita)
 - **Ganap** – buong salita ang inuulit (araw-araw, bahay-bahay)
- **Tambalan** – pinagsamang dalawang salita
 - **Malatambalan** – ang dalawang salitang pinagsama ay nananatili ang orihinal na kahulugan
(hal: bahay-kubo, silid-aralan)
- **Tambalang Ganap**
 - Pagsasama ng dalawang salita na nagbubuo ng bagong kahulugan
(Halimbawa: **bahaghari**)

Akademikong Sulatin

- **Abstrak** – maikling pagbubuod ng tesis o pag-aaral
- **Bionote** – impormatibong tala tungkol sa may-akda o tagapagsalita
- **Sintesis** – buod ng mga pangunahing ideya mula sa iba't ibang sanggunian
- **Book Report** – pagtalakay o pagsusuri sa kabuuan ng isang libro
- **Rebyu** – muling pagtingin at pagsusuri sa isang produkto, akda, o palabas

Batayang Kaalaman Sa Metodolohiya Ng Pananaliksik Panlipunan

- **SWOT Analysis** – pagsusuri sa lakas (Strengths), kahinaan (Weaknesses), oportunidad (Opportunities), at banta (Threats)
- **Etnograpiya** – pag-aaral sa kultura at kaugalian ng tao
- **Pananaliksik Leksikograpiko** – pagtatala at pag-aaral ng mga depinisyon
- **Video Documentation** – paggamit ng bidyo bilang tala ng pag-aaral o obserbasyon
- **Literature Review** – pagsasama-sama ng mga kaalaman mula sa mga naunang pag-aaral o akda
- **Pagtatanong/Obserbasyon/Panayam/Interbyu**
 - *Interbyu* – direktang pagtatanong sa isang eksperto o resource person
- **Participant Observation** – aktibong pagsubaybay sa gawain at kilos ng mga respondente
- **Pagsusuri ng Dokumento, Babasahin, at Panoorin** – pag-aaral ng mga nakasulat o naitalang materyales

- **Comparative Analysis** – paghahambing ng dalawang o higit pang datos o konsepto
- **Discourse Analysis** – pagsusuri sa gamit at pilosopiya ng wika sa komunikasyon
- **Content Analysis** – sistematikong pagsusuri ng mga dokumento, teksto, o mensahe

CONTEMPORARY WORLD • LIZWORIZ • RPH

Three Waves Of Migration Theory (D’AIM)

- **Henry Otley Beyer**
- **Negritos/Aetas** – Aboriginal settlers
- **Indones** – unang manlalayag, sea ferrers
- **Malay** – last, lived in barangay (30-100 families)

Core Population Theory

- **Evolution Theory** of F. Landa Jocano
- **Tabon Man** – oldest confirmed human, discovered by Robert Fox & Manuel Santiago in Tabon Cave, Palawan

Pre-Spanish Period

SOCIAL CLASS

- **Maharlika** – nobles (datu, rajah, etc.)
- **Timawa** – free man
- **Alipin Saguiguilir** – without own home
- **Alipin Namamahay** – has their own home

RELIGION

- **Paganism** – worship different Gods
- **Animism** – worship the environment (sun, moon, etc.)

ECONOMY

- Revolves with barter and trade
- First Traders:
 - **Orang Dampuans** – Champa Vietnam
 - **Banjasasins** – Indonesia
 - **Chinese** – Ming Dynasty

GOVERNMENT

- **Datu** – head of the state, pre-spanish president
- **Umalohokan** – barangay announcer, town crier
- **Trial by Ordeal** – form of punishment given by datu

COMMUNITIES

- **Igorot** – lived in cordilleras, practiced mummification
- **Pintados** – tattooed people in panay

Terms To Remember

- **Bothoan** – first school of pre-colonial filipinos
- **Babaylan** – doctors
- **Baybayin** – first system of writing
- **Sanduguan** – seal of friendship, blood compact
- **Bethrotal** – pre-hispanic engagement
- **Dowry** – originated in India, giving gift to partner
- **Makhdum/Merchanr** – arab missionary who brought Islam in Philippines in 1380

- **Sulu** – first sultanate by Abu Bakr in 1405
- **Tawi-tawi** – first settlement of Islam

EFIC – Internal Criticism: Content
 External Criticism: Form

Treaty Of Tordesillas (PEWS)

TREATY OF TORDESILLAS (PEWS)

- Divides the world into two
- Portugal East, West Spain

Social Classes (PIMPI)

- **Peninsulares** – Spanish born in spain
- **Insulares** – Spanish born in philippines
- **Mestizo** - Foreign
- **Principalia** – Middle class filipinos (ilustrados)
- **Indio** – Native Filipino

Spanish Policies

- **Pueblo** – town
- **Tribute** – tax
- **Cabeceras** – town center
- **Cabeza** de barangay – collector, brgy captain
- **Reduccion** – resettlement
- **Bandala** – product tax
- **Polo y servicio** – sapilitang paggawa
- **Polistas** – polo worker
- **Falla** – exemption tax for polo
- **Ayuntamiento** – trade center
- **Alcaldia** – areas pacified by Spanish
- **Corregimientos** – unpacified areas
- **Indulto de comercio** – privilege given to governors to engage in trade
- **Frailocracy** – government led by friars

Orders Of Spanish Expedition (MaLoCaSaViPi)

Ferdinand Magellan (1519)	Alvaro de Saavedra (1527)
Garcia Jofre Loaisa (1525)	Rudy Lopez de Villalobos (1542)
Sebastian Cabot (1526)	Miguel Lopez de Legazpi (1564)

Order Of Missionaries (AFJDRB)

Augustinians (1565)	Dominicans (1587)
Franciscans (1578)	Augustinian Recollects (1606)
Jesuits (1581)	Benedicts (1885)

Terms To Remember

- **St. Agustine** – prominence to God’s faith
- **St. Thomas Aquinas** - prominence to God’s reason
- **Gobernadorcillo** – pinakamataas na posisyon para sa mestizo
- **Governor-General** – direct govern PH
- **Council of Indies** – appoints gov gen

Treaty Of Paris

- Treaty wherein PH was sold to America
- 20 million Dollars as peace agreement

Magellan's Expedition (TSVSC)

- **March 16, 1521** – arrival in homonhon, samar
- Boats of Magellan with crew of **270 men**:
 - **Trinidad**
 - **Santiago**
 - **Victoria** – nakabalik
 - **San Antonio**
 - **Concepcion**
- **RAJAH KULAMBO** – first blood compact between Magellan and Kulambo in Limasawa
- **LEGASPI AND SIKATUNA** – first blood compact in Bohol
- **ENRIQUE OF MALACCA** – translator of Magellan
- **ANTONIO FIGAFETTA** – writer and secretary of Magellan

Year 1521

- **MARCH 31** – first mass in Limasawa
- **APRIL 7** – Magellan went to cebu
- **APRIL 14** – M baptize his wife
- **APRIL 21** – M plant cross
- **APRIL 27** – Magellan died by lapu-lapu (lower left leg)

Firsts To Remember

1. **Ventura Delos Reyes** – first filipino representative in Spanish cortes
2. **Dawn Man** – first to come in ph
3. **Ferdinand Magellan** – first to circumnavigate the earth
4. **El Cano** - first man to circumnavigate
5. **Doctrina Cristiana** – first book by Juan Dela Placencia
6. **Barlaan At Josaphat** – first filipino translated book by Antonio De Borja
7. **Miguel Lopez De Legazpi** – first gov gen in Spanish
8. **General Wesley Merritt** – first gov gen in America
9. **Thomasites** – first teachers
10. **Theodora Alonzo** – Rizal's first inspiration (Mi Primera Inspiracion)
11. **El Amor Patrio** – Rizal's first poem in Spanish
12. **Elpidio Rivera Quirino** – first Ilocano president
13. **Legaspi & Sikatuna** – blood compact in Bohol
14. **Diwata One** – first Philippine microsatellite
15. **Agustinians** – first missionary
16. **Lapu-Lapu** – first filipino hero
17. **Felipe Agoncillo** – first filipino diplomat
18. **Theodoro Agoncillo** – madamdaming manalaysay
19. **Cayetano & Arellano** – first chief justice
20. **Manuel Quezon** – first senate president of PH

21. **Deodato Arellano** – first president of Katipunan
22. **Ciudad del Santisimo Nombre de Jesus** – first town in PH
23. **Malolos Constitution** – first official constitution
24. **Pilipino Muna** – first filipino policy
25. **Kulambuan & Magellan** – first blood compact in PH
26. **Manila Bulletin** – oldest existing newspaper
27. **Letran** – oldest college in PH & second in Asia
28. **UST** – first university in PH
29. **Damian Domingo** – first PH painter
30. **Santiago de Vera** – first president of Royal Audencia

Spanish Governor-General

- **Miguel Lopez de Legaspi** – first
- **Diego delos Rios** – last
- **Jose Basco** – tobacco monopoly
- **Narciso Claveria** – surname for tax collection and census
- **Carlos Maria dela Torre** – best and most loved gov gen

Key Figures in PH Colonial History

- **Rafael Izquierdo** – execution of GOMBURZA
- **Eulogio Despujol** – ordered Rizal's arrest in Dapitan
- **Camilo Polavieja** – ordered Rizal's execution
- **Ramon Blanco** – declared 8 provinces in Martial Law
- **Manuel Blanco** – flora de filipinas (PH Botany)
- **Pedro Pilaez** – founder of secularization movement
- **Mariano Gomez** – confirmed secularization movement

Rise Of Nationalism

- **Secularization** – movement done by Pedro Pelaez, confirmed by Burgos
- **Cavite Munity** – resulted to execution of GOMBURZA
- **La Liga Filipina** – lasted 3 days, reason of Rizal's exile
- **Katipunan** – July 7, 1892, formed by Andres Bonifacio which resulted to Rizal's death
- **Tejeros Convention** – first election and resulted to Bonifacio's death.
- **Biak na Bato** - PH was sold by Katipunan for 800,000 Mexican Peso or 200M Dollars
- **Hong Kong** – where Emilio Aguinaldo was exciled
- **Propaganda Movement** – spearhead by ilustrados

Union Obrera Democratica

UNION OBRERA DEMOCRATICA

- By Isabelo Delos Reyes
- First labor union of the Philippines

The Triumverate

MARCELO HILARIO DEL PILAR

- Father of PH Journalism
- Father of PH Mansory
- **PEN NAMES:** Plaridel, Dolores Manapat, Piping Dilat at Siling Labuyo
- Diaryong Tagalog, Dasalan at Toksohan

GRACIANO LOPEZ JAENA

- First editor of La Solidaridad
- Great Orador
- **PEN NAME:** Diego Laura
- Fray Botod

JOSÉ RIZAL

- National Hero of the Philippines
- **PEN NAMES:** Laong Laan, Dimasalang, Agno
- Noli Me Tangere, El Filibusterismo, Sobre la Indolencia de los Filipinos

LA SOLIDARIDAD

- February 15, 1889
- newspaper of propaganda

LA LIGA FILIPINA

- July 3, 1892
- National socio-civic organization founded by Rizal

Kataastaasan Kagalang-galang Na Katipunan Ng Mga Anak Ng Bayan (KKK)

- July 7, 1892
- **Kalayaan** – newspaper of Katipunan
- **Kartilya** – guide book of Katipunan
- **Founder:** Andres Bonifacio
- **Objectives:** Political, Moral, Civil
- Katipon (Anak Bayan), Kawal (GomBurZa), Bayani (Rizal)

August 19, 1896

- **Teodoro Patino** – katipunerong nagsuplong sa Katipunan
- **Padre Mariano Gil** – paring nagsuplong sa Katipunan

August 23, 1896

- Sigaw sa pugad lawin sa bahay ni Melchora Aquino (Ina ng Katipunan)

December 30, 1896

- Death of Rizal ordered by **Polaveija**
- **“consummatum est”** means it is **finish**

JOSE PROTACIO RIZAL MERCADO Y ALONZO REALONDA

- **Born:** June 19, 1961, in Calamba, Laguna
- **Died:** December 30, 1896
- **Parents:** Theodora Alonzo and Francisco Mercado (Don Kikoy)

- **Pets:** Usman (Dog) and Alipato (Horse)
- **Nicknames:** UTE (Siblings) and PEPE (Town)
- **Schools Attended:**
 - Private School in Binan
 - HS: Ateneo
 - College: UST
- **Greatest Love:** Leonor Rivera (Maria Clara)
- **Baptize:** Rufino Collantes

Works Of Rizal

- **RA 1425** – LIZWORIZ in curriculum
- **Ala Juventud Filipina** – greatest crowning glory in UST
- **Mi Primera Inspiracion** – written in Dapitan
- **Flowers of Heidelberg** – rizal was reminded by his hometown while in Germany
- **Ricial** – means wheat field
- **SPECIES (ABDFRT)**
 - Abogania Bettie (salagubang)
 - Draco (flying lizard)
 - Rhacophorus Frog (flying frog)

NOLI ME TANGERE

- Berlin, Germany in 1887
- Touch Me Not/ The Social Cancer
- Inspired by Uncle Tom's Cabin by Harriet Beecher Stowe
- About: Love, Motherland, Ibarra, Social Cancer
- Funded by Maximo Viola (2000 copies)

EL FILIBUSTERISMO

- Ghent, Belgium in 1891
- The Reign of Greed
- Inspired by Count of Monte Cristo by Alexander Dumas
- About: GOMBURZA, Simeon
- Funded by Valentin Ventura

Rizal's Biographers

- **Biografia de Rizal** – Rafael Palma
- **The Great Malayan** – Carlos Quirino
- **Jose Rizal: Life and Times** – Camilo Osias
- **Jose Rizal: Life, Works, and Writing** – Gregorio F. Zaide
- **The First Filipino: A Biography of Jose Rizal** – Leon M. Guerrero

Rizal's Life

- **Leon Monroy** – Rizal learned Spanish
- **Juancho** – Rizal learned painting skills
- **Age of 3** – Rizal learned the alphabet
- **Age of 5** – Rizal learned to read and write
- **Tio Manuel** – Rizal learned wrestling
- **Jose Bech** – Rizal's first teacher in ATENEO
- **Circulo Hispano Filipino** – association formed by Filipino students in Spain which composed of Spaniards and Filipino reforms
- **Little But Expensive** – last article of La Solidaridad on November 15, 1895
- **Cadiz March** – sang during Rizal's execution
- **97 and 36** – lotto number where Rizal won 6, 200 (father: 2000, friend: 200)

Discovered After Dr. Jose Rizal

- **Rachophorus Rizalii** - Rizal's frog.
- **Rhinoceros Rizalii** - Rizal's rhinoceros beetle.
- **Kandul Rizalii** - Rizal's catfish.
- **Parantica Rizalii** - Rizal's butterfly.
- **Cardiodactylus Rizalii** - Rizal's cricket.
- **Pachyrhynchus Rizalii** - Rizal's beetle.
- **Hydrozoa Rizalii** - Rizal's jellyfish.

June 12, 1898

- Araw ng kasarinlan
- Singing of the Marcha Nacional Filipina
- **Band:** SFactsan Francisco de Malabon Band
- **Lyrics:** Jose Palma
- **Music:** Julian Felipe

Other Facts

- **CARINOSA** - means affectionate one and character associated with Filipino women
- **APARTHEID** - Separation of Black and White people
- **MOCK BATTLE OF MANILA BAY** – end of Spanish regime (Spanish-American war)

American Intervention

- **MAY 1, 1898** – Manifest destiny
- **DEC. 10, 1898** – Benevolent Assimilation by William McKinley

Philippine-American War

- **FEBRUARY 4, 1899** - San Juan Bridge Shooting Incident

Other Facts

- **JUNE 5, 1899** – death of Luna, killed by Filipino
- **DECEMBER 2, 1899** – death of Goyo
- **MARCH 22, 1901** – aguinaldo was captured in Palanan, Isabela

Traitors Of The Era

- Katipunan – Theodoro Patino
- Pasong Tirad – Juanario Galod
- Makabebe Scouts – emilio arrest in isabela
- Makapili – Japanese reign

American Period (MCC)

Military (MOA)

- Wesley Merritt, Elwell C. Otis, MacArthur Douglass (first and only field marshall)

Civil

- William Howard Taft, Frank Murphy

Commonwealth

- Filipino government

Terms To Remember

- **Payne-Aldrich Tariff Act** – partial free trade (quota)
- **Simmon-Underwood Act** – full free trade

- **Tydings McDuffie Law** – 10-year transition or Philippines Independence Act: from Quezon
- **Hare Hawes Cutting Act** – from Osmena and Roxas
- **Gabaldon Act** – building of modern public schools

Bell Trade Act

- **PARITY RIGHTS** - American rights on natural resources
- Less to no tariffs on imports from US
- Fixed exchange rate of peso to dollar (2:1)

Jone's Law

- **Bicameralism** – 2 chambers: Lower and Upper House
- Grant of independence as soon as a stable government can be established.
- **Preamble of Jones Law** – independence

Remarkable Women

- **Gregoria Montoya** – Joan of Arc in Cavite
- **Magbanua** – Joan of Arc in Visayas
- **Kahabagan** – Joan of Arc in Tagalogs
- **Tecson** – Red Cross, Girls Scout
- **Josephina Guerrero** – lepers spy
- **Gabriela Silang** - first female leader of the revolutionary movement against the Spanish colonizers.
- **Goryang** – may bahay ni Bonifacio
- **Maria Clara** – greatest love of Rizal (Leonor Rivera)
- **Marcela Agoncillo** – wife of emilio Aguinaldo, sew the ph flag
- **Cory Aquino** – first female president of PH and Asia, Mother of Democracy
- **Melchora Aquino** – mother of Katipunan (Tandang Sora)
- **Teodora Alonzo** – woman of conviction
- **Sisa** – symbolizes filipino abused women

Other Facts

- **ANTONIO LUNA** - bravest general of the PH during the American Period
- **MARIANO SAKAY** – president of katagalogan
- **JAPANESE** – golden age in History of PH Literature

Japanese Period

- **Educational Focus:** Tech Voc/Love for Work and Dignity for Labor
- **AUGUST 6** – Little Boy was drop in Hiroshima
- **Hukbalahap** – Hukbong Bayan Laban sa Hapon
- **Makapili** – pro-japanese
- **Kempeitai** – Japanese police
- **Haiku** – 3 lines, nature, 575, 17 syllabuses
- **Matsuo Bashō** – greatest haiku writer
- **Tanka** – 5 lines, nature and love, 57577, 34 syllabuses

Filipino Pen Names Of Prominent Writers	
Pangalan ng Manunulat	Sagisag-Panulat / Pen Name
Amado Hernandez	Amante Ernani, Herminia de la Riva, Julio Abril
Andres Bonifacio	Agapito, Bagumbayan, Maypagasa, Magdiwang, Great Plebian, Anak Bayan
Antonio Luna	Taga-ilog
Apolinario Mabini	Bini, Paralitiko, Katabay
Cecilio Apostol	Catulo, Calipso, Calypso
Emilio Aguinaldo	Rosalia Magdalo, Magdalo
Emilio Jacinto	Dimas-ilaw, Pingkian, Brain of Katipunan
Epifanio delos Santos	G. Solon
Felipe Calderon	Simoun, Elias
Fernando Ma. Guerrero	Fluvio Gil, Florisel
Florentino Collantes	Kuntil-butil
Francisco Baltazar	Balagtas
Graciano Lopez Jaena	Bolivar, Diego Laura
Jose Corazon de Jesus	Huseng Batute, Pusong Hapis, Luksang Paruparo
Jose dela Cruz	Huseng Sisiw
Jose Garcia Villa	Doveglion
Jose Maria Panganiban	Jomapa, J.M.P.
Jose Palma	Anahaw, Esteban Estebanes, Gan Hantik
Jose Rizal	Dimasalang, Laong Laan, Agno
Juan Luna	J.B., Buan
Lope K. Santos	Anak-bayan, Doktor Lukas, Lakandalita
Marcelo H. del Pilar	Plaridel, Dolores Manapat, Piping Dilat, Siling Labuyo
Mariano Ponce	Naning, Tikbalang, Kalipulako
N.V.M. Gonzalez	N.V.M. Gonzalez

Nick Joaquin	Quijano de Manila
Pascual H. Poblete	Anak-bayan
Pio Valenzuela	Madlang-awa
Severino Reyes	Lola Basyang
Virgilio Almario	Rio Alma

Presidents Of The Philippines	
Name	Information
Emilio Aguinaldo 1899-1901	•1st pres of 1st republic •First Luzon Pres •Revolutionary Government •Youngest pres. 28 yrs old •Longest live 94 yrs old •Only pres of 1st republic •VP-Mariano Trias
Manuel L. Quezon 1935-1944	•1st pres of Commonwealth •Father of National Language •Tuberculosis death •Aurora Quezon •Aurora wife •New York dead Father of Social Justice
José P. Laurel 1943-1945	•Only pres. 2nd republic •Puppet President •Lyceum of the Philippines
Sergio Osmeña 1944-1946	•2nd pres. of Commonwealth •1st Visayan president •VP ni Quezon •Shortest serving president •First Beeker of PH Assembly
Manuel Roxas 1946-1948	•1st pres. of 3rd republic •Gave amnesty to arrested Filipino who connived with Japanese
Elpidio Quirino 1948-1953	•2nd pres of 3rd republic •First Ilocano Pres •Integrity Board •Hydro Electric Power Plant •Golden Arinola •Created social security commission
Ramon Magsaysay 1953-1957	•3rd pres of 3rd republic •Man of the masses •Little man •Golden Years (no corrupt) •Wear slipper, barong tagalog •Plane crash death •From Zambales
Carlos P. Garcia 1957-1961	•4th pres of 3rd republic •Filipino First Policy •Austerity Program (Traded Industry) •1st president buried in Libingan ng mga bayani •Philippine Prosperity Program
Diosdado Macapagal 1961-1965	•5th pres of 3rd republic •Zero corruption /Incorruptible president

	•Land for the landless •Tenacy Act (75 Hectares retention limit) •Comprehensive Agrarian Reform Act (CARE) •Veterans Bank •Landbank •Nagbalik ng Kalayaan July 4-June 12 •Sertipiko sa Filipino
Ferdinand Marcos 1965-1986	•4th republic •Infrastructure Man: SLEX, NLEX, PGH, PICC etc. •Miracle Rice •#1 exporting country (Vietnam) •“I will make this country great again” • 7 Hectares retention limit
Corazon Aquino 1986-1992	•1st female president •1st pres 5th republic •Mother of Democracy •Family planning •Restored democracy •1987 constitution
Fidel V. Ramos 1992-1998	•2nd pres of 5th republic •Father of Democracy •Father of Army •Philippines 2000
Joseph Estrada 1998-2001	•3rd pres of 5th republic • Father of the masses •Presidential Commission on Educational Reform •Baba sa Pwesto Edsa Revolution 2001
Gloria Macapagal Arroyo 2001-2010	•4th pres of 5th republic •People of the Philippines •Holiday Economic •High GDP •Evat •Strong republic
Benigno Aquino III 2010-2016	•5th pres of 5th republic •K12 •No Wang wang Policy •Tuwid na Daan •First bachelor president
Rodrigo Duterte 2016-2022	•6th pres of 5th republic •First Mindanaon president •Extrajudicial killing issue •Build Build Build •Federalism •Human rights issue
Bongbong Marcos 2022-present	•7th pres of 5th republic(17th) •Elected 2022 •Unity platform •Son of Ferdinand Marcos •Free trade expansion •Kadiwa program boost •Maharlika Investment Fund

- **Malolos/ 1st Republic:** 1 (Aguinaldo)
- **Commonwealth:** 2 (Quezon, Osmena)
- **2nd Republic (Japanese):** 1 (Laurel)

- **3rd Republic:** 5 (Roxas, Quirino, Magsaysay, Garcia, Macapagal) ROQUIMAGARMA
- **Bagong Lipunan/ 4th Republic:** 1 (Marcos)
- **5th Republic:** 7 (Aquino, Ramos, Estrada, Arroyo, Aquino, Duterte, BBM)

MANUEL L. QUEZON – National Land Settlement Administration (NLSA)

ELPIDO QUIRINO – Land Settlement and Development Corporation (LASEDECO)

RAMON MAGSAYSAY - National Resettlement and Rehabilitation Administration (NARRA)

CORY AQUINO - Comprehensive Agrarian Reform Law (CARL)

Mga Pilipinong Bayani at Kanilang Titulong Pangkasaysayan	
Pangalan	Kilalang Titulo / Ambag
Jose Rizal	Pambansang Bayani , may-akda ng Noli Me Tangere at El Filibusterismo
Andres Bonifacio	Ama ng Rebolusyong Pilipino/Ama mg Himagsikan/ Sumpremo ng Katipunan , lider ng Katipunan
Emilio Aguinaldo	Unang Pangulo ng Pilipinas , namuno sa laban kontra Espanya at Amerikano
Emilio Jacinto	Utak ng Katipunan , sumulat ng Kartilya ng Katipunan
Apolinario Mabini	Dakilang Lumpo/ Utak ng Rebolusyon , unang Punong Ministro ng Pilipinas
Marcelo H. del Pilar	Dakilang Propagandista/ Ama ng Pahayagang Tagalog , editor ng La Solidaridad
Graciano Lopez Jaena	Tagapagtatag at unang editor ng La Solidaridad , mahusay na orador
Antonio Luna	Heneral ng Hukbong Pilipino , kilala sa disiplina at katalinuhan
Melchora Aquino	Kilala bilang Tandang Sora/ Ina ng Katipunan , inang tagapag-alaga ng Katipunero

Juan Luna	Alagad ng sining at aktibong propagandista , Spolarium , The Blood Compact , Death of Cleopatra
Gregorio del Pilar	Bayani ng Tirad Pass/ Boy General , pinakabatang heneral ng rebolusyon
Diego Silang	Tagapagpalaya ng Ilocos , lider ng pag-aalsa sa Ilocos laban sa Kastila
Gabriela Silang	Joan of Arc ng Ilocos , Unang babaeng pinuno ng pag-aalsa, asawa ni Diego Silang
Francisco Dagohoy	Pinuno ng pinakamatagal na rebelyon sa Bohol (85 taon)
Sultan Kudarat	Pinuno ng Muslim Mindanao , lumaban sa kolonyalismong Espanyol
Macario Sakay	Huling Heneral ng Himagsikan/Katipunan , nagtatag ng Republika ng Katagalugan , lumaban sa Amerikano
Trinidad Tecson	Ina ng Biak na Bata , Ina ng Red Cross , Katipunera at rebolusyonaryo
Julian Felipe	Kompositor ng Lupang Hinirang , Pambansang Awit
Pio Valenzuela	Doktor at pinuno ng Katipunan, nag-ulat kay Rizal sa Dapitan
Pio del Pilar	Heneral mula Makati , lider ng Katipunan sa timog Maynila
Jose Abad Santos	Chief Justice of Supreme Court (1941) , Punong Mahistrado , pinatay ng Hapon dahil sa katapatan sa bayan <i>"It is an honor to die for one's country. Not everybody has that chance."</i>
Vicente Lim	Unang Pilipino sa West Point, bayani ng Ikalawang Digmaang Pandaigdig
Josefa Llanes Escoda	Tagapagtatag ng Girl Scouts of the Philippines , martir ng Bataan

Carlos P. Romulo	Pangulo ng UN General Assembly , diplomat at manunulat
Lapu-Lapu	Unang bayani ng Pilipinas , lumaban kay Ferdinand Magellan sa Mactan
Corazon Aquino	Unang babaeng pangulo ng Pilipinas, simbolo ng People Power Revolution
Manuel L. Quezon	Ama ng Wikang Pambansa , pangulo ng Commonwealth
Ramon Magsaysay	Idolo ng Masa , pangulo na laban sa korapsyon at komunismo
Ka Lorenzo Tañada	Makabayang senador , tagapagtanggol ng karapatang pantao
Wenceslao Vinzons	Bayani ng Camarines Norte , pinatay ng Hapon noong WWII

Important Terms

- **Geneva Convention** – to cure wounded soldiers
- **Kyoto Protocol** – mitigating global warming
- **Montreal Protocol** – mitigating ozone layer depletion
- **Philippine Clean Air Act of 1999** – ban the use of incinerators

World Civilization

Ancient Civilization

- **Tigris and Euphrates** – Fertile Crescent in Sumerian Mesopotamia
- **Huang He-Yangtze (Yellow River)** – China
- **Nile River** – Egypt
- **Indus River** – India
- **Aegean Sea** – Greece

System Of Writing

- **Cuneiform** –Mesopotamia
- **Hanzi/ Bone Script** – China
- **Hieroglyphics** – Egypt
- **Indus Script** – India
- **Linear B** – Greece
- **Baybayin/Alibata** – Philippines

Religion

- **Hinduism** – oldest existing religion in Asia
- **Optimism** - Eastern region
- **Mysticism** – western region, believed in one God
- **Buddhism** – focuses on ending suffering through enlightenment

- **Islam** – monotheistic religion, believes in Allah and the prophet Muhammad
- **Christianity** – based on the teachings of Jesus Christ
- **Judaism** – oldest monotheistic religion, based on the Torah
- **Taoism** – living in harmony with the Tao (the Way)
- **Confucianism** – teaches moral values and respect for elders
- **Shintoism** – traditional religion of Japan, nature and ancestor worship
- **Zoroastrianism** – ancient Persian religion, battle between good and evil
- **Sikhism** – belief in one God, equality, and service to others
- **Animism** – belief that spirits exist in nature (trees, animals, etc.)
- **Atheism** – belief in no god or divine being
- **Agnosticism** – unsure if God exists or not
- **Polytheism** – belief in many gods
- **Monotheism** – belief in one God
- **Reincarnation** – belief in rebirth after death (common in Hinduism & Buddhism)
- **Nirvana** – state of liberation from suffering (Buddhism)
- **Karma** – the idea that actions affect future outcomes (Hinduism, Buddhism)

Elements of State (PSGT)

- **Population**
- **Sovereignty** – freedom from foreign control
- **Government**
- **Territory** – area of land, sea, space of a country

Inherent Powers Of The State

- **Power of Expropriation** – power of emanate domain: public use
- **Police Power** – general welfare
- **Taxation** – charge imposed
 - **Proportion** – constant
 - **Progressive** – high income & tax
 - **Regressive** – high income, low tax

Form Of Government

FORM OF GOVERNMENT

- **Anarchy** – no government
- **Monarchy** – king or queen
- **Aristocracy** – Elites
- **Oligarchy** – Few
- **Democracy** – People
- **Parliamentary** – controls legislative and executive
- **Federalism** – national to national
- **Necrocracy** – local to local
- **Meritocracy** – people lead
- **Frailocracy** – led by friars
- **Communism** – no private property
- **Modern Capitalism** – entrepreneur

Other Important Terms

- **DEMOCRATIC REPUBLIC** – Philippine’s form of government
- **CONGRESS** – power to declare the existence of War
- **HOUSE OF REPRESENTATIVE** – imposed impeachments
- **SENATE** – trial and decide impeachment

Industrial Revolution Stages

Industrial Revolution Stages

1. **First Industrial Revolution** - Early industries and transportation.
2. **Second Industrial Revolution** - Mechanized production and mass production.
3. **Third Industrial Revolution** - Computer & Internet - Marked the digital age.
 - **Father of Computer** - Charles Babbage.
 - **Father of Modern Computer** - Alan Turing.
 - **Mother of Computer** - Ada Lovelace.
4. **Fourth Industrial Revolution** - Automation and smart machines.

Population Control

- **Abstinence** - The most effective way to prevent population growth.

Largest & Smallest Countries

- **Largest Countries in Asia**
 1. **Kazakhstan** - Central Asia.
 2. **China** - East Asia.
 3. **India** - South Asia.
 4. **Saudi Arabia** - West Asia (Middle East).
 5. **Indonesia** - Southeast Asia.
 6. **Pakistan** - South Asia.
 7. **Iran** - West Asia (Middle East).
- **Smallest Country in the World** - Vatican City (Rome, Italy).

Largest & Smallest Provinces in the Philippines (PALP-BANA)

- **Palawan** - Largest province in the Philippines.
- **Batanes** - Smallest province in the Philippines.

Geological Periods of the Earth (ProMAMCE)

1. **Precambrian** - Earliest life forms.
2. **Paleozoic** - Marine life.
3. **Mesozoic** - Age of dinosaurs.
4. **Cenozoic** - Age of mammals.
5. **Quaternary** - Modern human appearance.

Historiography

- Study of how history is written.
- Includes how different writers interpret historical events.
- History has both **objective** and **subjective** aspects.

Factors Affecting Historiography

- Time it was written
- Writer’s bias
- Writer’s intention
- Proximity to the event

Philosophy Of History	
Thinker	Core Concepts & Highlights
Herodotus	- Father of History - Wrote “ The Histories ” (Greco-Persian Wars) - Used narrative and descriptive style
Thucydides	- Wrote “ History of the Peloponnesian War ” - Scientific, analytical, and impartial - Focused on cause and effect
Giambattista Vico	- History as product of human nature - Emphasized ideas, customs, deeds - Introduced universal history
G.W.F. Hegel	- History has a telos (goal) - Introduced Dialectics : Thesis → Antithesis → Synthesis - Movement toward human freedom
Karl Marx	- Developed Materialist Dialectics - History driven by economic forces - History as class struggle toward communism
Fernand Braudel	- Founded Annales School - Emphasized economic, social, cultural context - Importance of long-term historical structures
Michel Foucault	- Concept of “ Archeology of Knowledge ” - Denied objective truth - History as power relations and discourse
Zeus Salazar	- Promoted Pantayong Pananaw - Shared cultural meanings within a group - History must be understood from within the Filipino context

Historical Sources

Primary Sources

- First-hand accounts (e.g., diaries, photos, letters)
- Written during the event

Secondary Sources

- Interpretations of primary sources
- Written after the event

Elements Of History

1. People
2. Date
3. Place
4. Significance

Controversies in Philippine History

First Mass in the Philippines

- **Ano 'to?:** Debate tungkol sa **saan talaga ginanap ang unang misa** sa Pilipinas.
- **Bakit kontrobersyal?:** May **magkaibang account** kung saan nangyari — sa **Mazaua** o **Butuan**.
- **Ano'ng nangyari?**
 - ✓ **Antonio Pigafetta** (chronicler of Magellan): Sinabing naganap ang misa sa **Mazaua** (near **Humonhon**, Eastern Visayas).
 - ✓ **Fr. Urdaneta:** Sinasabing nagtangkang tumigil ang mga Kastila sa **Camiguin at Masava**, pero napadpad sa **Butuan**, tapos sa **Bohol** dahil sa agos ng dagat.
- **Bottom line:** Wala pang final proof kung **Mazaua** o **Butuan**, pero **Limasawa** sa Southern Leyte ang opisyal na kinikilala sa kasalukuyan (ng NHCP).

Cavite Mutiny (1872)

- **Ano 'to?:** Rebellion ng ilang Pilipinong sundalo at manggagawa sa **Fort San Felipe, Cavite**, laban sa mga Kastila.
- **Bakit kontrobersyal?:** **Magkaibang pananaw** kung bakit nangyari ang mutiny at kung sino ang sangkot.
- **Ano'ng nangyari?**
 - ✓ **Jose Montero Vidal** (Spanish writer): Sabi niya, planong **asasinuhin ang mga opisyal**, pero **napagkamalan lang ang fireworks** sa Sampaloc fiesta bilang signal ng pag-aalsa.
 - ✓ **Gen. Rafael Izquierdo** (Spanish governor): **Isinisi sa mga paring Pilipino** at layunin daw nitong **pabagsakin ang pamahalaan**.
 - ✓ **Dr. Trinidad Pardo de Tavera** (Filipino scholar): Maliit lang daw ang mutiny at **ginamit lang ito ng Kastila at mga prayle** para mapatahimik ang mga

Pilipinong repormista, gaya nina GomBurZa, na pinatay pagkatapos nito.

- **Bottom line:** Ang mutiny ay naging dahilan ng pagbitay sa GomBurZa, na nag-udyok sa kilusang propaganda at sa diwang makabayan ng mga Pilipino.

Cry of Pugad Lawin (1896)

- **Ano 'to?:** Simbolo ng simula ng himagsikan laban sa Espanya, kung saan pinunit ng mga Katipunero ang kanilang mga cedula (residence certificate) bilang pagsuway sa kolonyal na pamahalaan.
- **Bakit kontrobersyal?:** Iba-iba ang accounts sa kung kailan at saan ito nangyari.
- **Ano'ng nangyari?**
 - ✓ **Dr. Pio Valenzuela:** Nangyari ito sa Aug. 22–23, 1896, sa bahay ni Juan Ramos (anak ni Melchora Aquino).
 - ✓ **Mariano Alvarez:** Aug. 23–24, 1896 sa Sampaukan, Bahay Toro, may higit 1000 Katipuneros.
 - ✓ **Gregoria de Jesus:** Sinabi niyang Aug. 25 ang Cry at kinailangang tumakas siya mula sa mga sundalong Kastila.
 - ✓ **Guillermo Masangkay:** Aug. 26 sa Balintawak, pinangunahan ni Andres Bonifacio.
- **Bottom line:** Bagamat hindi sigurado ang exact date and location, malinaw na ito ang hudyat ng rebolusyon laban sa Espanya.

Oral History

- **Nature:** Unwritten, passed by memory
- **Danger:** Influenced by interviewer's background
- **Interpretation:** Must be critically evaluated and cross-checked

Local History

- Study of the past in a local context
- Helps build national identity and history

State

Definition (De Leon & De Leon, 2014)

A community with:

- People
- Territory
- Government
- Sovereignty

Powers of the State

- Eminent Domain
- Taxation
- Police Power

Constitutions

A. Malolos Constitution (1899)

- **Chairman:** Felipe Calderon | President: Emilio Aguinaldo
- Unicameral legislature, 4-year term
- **Suffrage:** Male, 21+, educated, ₱500 taxpayer
- **Key Provisions:**
 - ✓ Legalizes Independence
 - ✓ Separation of Church and State
 - ✓ Executive Committee replaces President temporarily

B. 1935 Constitution

- **Chairman:** Claro M. Recto | President: Manuel L. Quezon
- Created due to Tydings-McDuffie Law
- Formed Commonwealth Gov't
- Public debt limit set by U.S. Congress

C. 1973 Constitution

- **Chairman & President:** Ferdinand Marcos
- Parliamentary, unicameral, no presidential reelection
- **Notable:**
 - ✓ Voting age: 18+
 - ✓ No property/literacy requirement to vote
 - ✓ PM can dissolve legislature
 - ✓ Martial Law powers retained

D. 1987 Constitution

- **Chairman:** Cecilia Muñoz-Palma | President: Corazon Aquino
- Republican, bicameral, 6-year term, no reelection
- **Highlights:**
 - ✓ Independent foreign policy
 - ✓ No foreign military bases
 - ✓ Suspension of habeas corpus/martial law max 60 days
 - ✓ Creation of CHR (Commission on Human Rights)

Land Reform Laws

RA 1199 – Agricultural Tenancy Act of 1954

- President: Ramon Magsaysay
- Allowed leasehold system
- Tenants couldn't be ejected without just cause

RA 1400 – Land Reform Act of 1955

- Expropriated land >300 hectares
- Focused on areas with agrarian unrest

RA 3844 – Land Reform Code (1963)

- President: Diosdado Macapagal
- Land purchase for landless tenants
- Retention limit: 75 hectares
- Bill of rights for workers

Agricultural Land Reform Code of 1971

- President: Ferdinand Marcos
- Abolished personal cultivation as ejection reason
- Created Department of Agrarian Reform
- Converted share tenants to leaseholders

RA 6657 – Comprehensive Agrarian Reform Program (CARP)

- Max land retention: 5 hectares per owner
- 3 hectares can be given per child
- Based on land viability (terrain, commodity, etc.)

Taxation Theories		
Theory	Description	Key Idea
Ability-to-Pay	Tax based on a person’s income or capacity to pay	Those who earn more pay more tax
Benefit Principle	Tax paid should be proportional to the services received	Taxers pay according to government benefits

Types Of Taxation		
Type	Description	Effect
Progressive	Higher income → higher tax rate	Wealthy pay larger % of income
Regressive	Higher income → lower tax rate	Lower-income bear heavier tax burden
Proportional	Flat tax rate regardless of income	Everyone pays same % of income

TRAIN LAW (Tax Reform for Acceleration and Inclusion)	
Provision	Description
Simplifies tax system	Streamlined tax rates and brackets
Income below ₱250,000 annually	Exempt from paying income tax

Keyword	Fact
Globalization	A complex process of interconnectedness economically, politically, and culturally.
Arjun Appadurai	Anthropologist who identified five dimensions of globalization: ethnoscap es, mediascap es, technoscap es, financescap es, ideoscap es.
Neoliberalism	An economic model favoring free-market capitalism, deregulation, and reduction in government spending.
Bretton Woods System	Post-WWII agreement that created the IMF and World Bank in 1944.
Washington Consensus	A set of 10 economic policies promoting market liberalization in developing nations.
Transnationalism	Economic, political, and cultural processes that transcend national borders.
Deterritorialization	The weakening of cultural ties to geographic locations due to globalization.
Glocalization	Global and local interaction resulting in unique outcomes (e.g., McDonald's rice meals in Asia).
Cultural Imperialism	Dominance of one culture over others, often through media or economic power.
Internationalism	The principle of cooperation among nations for common good.
Nation-State	A sovereign state whose citizens are relatively homogeneous in culture or language.
Sovereignty	The authority of a state to govern itself without external interference.

Supranational Organization	An entity formed by multiple countries for collective goals (e.g., EU, UN).
Regionalism	Emphasis on regional rather than global cooperation (e.g., ASEAN).
ASEAN Economic Community	Launched in 2015 to create a single market and production base in Southeast Asia.
Cosmopolitanism	Belief in being a global citizen with moral obligations beyond borders.
Diaspora	Dispersion of a population from their original homeland.
Remittances	Money sent home by migrant workers, vital to many developing economies.
Global Citizenship	The idea of being a member of the world community with shared responsibilities.
Global South	Term referring to poorer, less developed countries, mostly in the Southern Hemisphere.
Global North	Term for wealthy, industrialized countries often in the Northern Hemisphere.
Migration	The movement of people across borders, voluntary or forced.
Refugee	A person forced to flee their country due to war, persecution, or disaster.
Statelessness	Condition of not being recognized as a citizen by any country.
Xenophobia	Fear or hatred of foreigners or strangers.
International Labor Migration	Movement of workers across countries in search of employment.
Fair Trade	Movement ensuring producers in developing countries get fair prices.

Outsourcing	Hiring of external organizations to perform business tasks abroad.
TNCs (Transnational Corporations)	Companies operating in multiple countries (e.g., Nestlé, Apple, Unilever).
WTO (World Trade Organization)	Regulates international trade between nations.
Protectionism	Economic policy of restricting imports to protect local industries.
Tariffs	Taxes imposed on imported goods.
Economic Globalization	Increasing economic interdependence of nations through trade and finance.
Cultural Globalization	The spread of cultural beliefs and norms across the globe.
Political Globalization	Growth of worldwide political standards and institutions (e.g., UN, ICC).
Environmental Globalization	Global cooperation and issues regarding the environment and climate change.
Climate Refugee	People forced to move due to environmental factors like rising sea levels.
Kyoto Protocol	1997 international treaty to reduce greenhouse gas emissions.
Paris Agreement	2015 climate accord to limit global warming below 2°C.
Sustainable Development	Development that meets present needs without compromising future generations.
SDGs (Sustainable Dev Goals)	17 UN goals to end poverty, protect the planet, and ensure peace and prosperity.
Digital Divide	The gap between people with and without access to digital technology.
E-commerce	Buying and selling products via the internet.

Cybersecurity	Protection of computer systems and data from digital attacks.
Cryptocurrency	Digital currency secured by cryptography (e.g., Bitcoin).
Social Media Activism	Use of social platforms for political or social change.
Global Protest Movements	Worldwide uprisings like Arab Spring or Black Lives Matter.
Disinformation	False information spread deliberately to mislead.
Media Literacy	The ability to critically analyze media content and sources.
McDonaldization	Process where global cultures adopt traits of fast-food chains (efficiency, uniformity).
Cultural Hybridization	Mixing of cultures to form new, blended ones.
Localization	Adapting global products to fit local cultures.
Global Governance	Political cooperation among countries to solve international issues.
G7 and G20	Groups of powerful economies that discuss and coordinate global policies.
UN Security Council	Most powerful UN body, responsible for maintaining international peace.
ICC (Intl. Criminal Court)	Tries individuals for crimes like genocide and war crimes.
IMF (Intl. Monetary Fund)	Provides financial assistance to struggling countries.
World Bank	Offers loans and grants to promote development in poor nations.
NGO (Non-Governmental Org)	Non-profit groups working on global issues like health, education, rights.
Global Risk	Events or trends that pose major threats to global stability (e.g., pandemics, cyberattacks).

Pandemic	A global disease outbreak (e.g., COVID-19) affecting multiple continents.
Vaccine Nationalism	When countries prioritize their own access to vaccines over global equity.
Post-Truth	Situation where emotions matter more than facts in shaping public opinion.
Global Inequality	Unequal distribution of wealth, opportunities, and access among nations.
BRICS	Group of emerging economies: Brazil, Russia, India, China, South Africa.

Theory	Direct Description
World Systems Theory	Immanuel Wallerstein ; world divided into core , semi-periphery , and periphery nations.
McDonaldization	George Ritzer ; global culture becomes efficient , predictable , and controlled like fast food.
Global Village	Marshall McLuhan ; the world feels smaller due to instant global communication .
Network Society	Manuel Castells ; society shaped by digital networks and global information flow .
Global Capitalism Theory	A transnational capitalist class dominates the global economy beyond national control.
Modernization Theory	Developing countries should follow Western industrial models to progress.
Dependency Theory	Poor countries stay poor due to exploitation by rich nations; a critique of modernization.

Deterritorialization	Activities like trade and communication happen beyond national borders .
Clash of Civilizations	Samuel Huntington ; future conflicts will be cultural , not just political or economic.
Global Governance Theory	Global institutions (e.g., UN, WTO) manage international rules and issues .
Cosmopolitanism	David Held ; promotes global citizenship and universal human rights .
Glocalization	Global products are adapted to fit local cultures (e.g., McDonald's local menus).
Time-Space Compression	Technology makes distant places feel closer and interactions faster .
Cultural Imperialism	Dominant cultures (like the U.S.) influence or suppress others through media and brands .
Liquid Modernity	Zygmunt Bauman ; society is unstable and fast-changing due to globalization.
Neo-Liberalism	Promotes free markets , privatization , and deregulation ; key in globalization.
Spaces of Flows	Castells ; global cities are hubs for digital, economic , and cultural exchanges .
Scapes Theory	Arjun Appadurai ; globalization flows in ethnoscape, technoscape, financescape , etc.
Hybridization	Cultures blend due to globalization (e.g., K-pop mixing with Western music).
Flattening of the World	Thomas Friedman ; tech and trade have made the global playing field more equal .

UNDERSTANDING THE SELF		
The Self from Philosophical Perspective		
Philosopher	Key Idea	Explanation
Socrates	"Know thyself"	The soul is the true self , not the body. The body is mortal , the soul is immortal . Life without self-examination is not worth living.
Plato	Tripartite Soul	The self is an immortal soul inside a mortal body . The soul has 3 parts: rational, spirited , and appetitive .
Aristotle	Golden Mean	The self is made of body and soul . Happiness comes from moderation and living with reason .
St. Augustine	Love for God and others	The soul is united with the body . True happiness is found in God . Avoid love of worldly things.
St. Thomas Aquinas	Reason and Divine Revelation	The self is guided by reason and faith . The soul seeks truth and perfection in God .
Descartes	"I think, therefore I am"	The self is the mind (immaterial) , separate from the body (material) . The mind is the center of the self.
John Locke	Tabula Rasa	At birth, the mind is a blank slate . Self is formed through experience . Individuals are free to shape themselves.
Immanuel Kant	Categorical Imperative	The self has inner (rational) and outer (physical) parts. Do actions you can wish to become a universal law .

Maurice Merleau-Ponty	Embodied Self	The self is a union of body and mind . Experiences happen through the body .
Gilbert Ryle	Mind-Body Unity	The mind is not separate from the body. Knowing is both "knowing how" and "knowing that" .
Paul Churchland	Materialism	The brain , not the mind, forms the self. Mental processes are biological and based on neural networks .

Western vs Oriental Views of the Self		
View	Key Term	Definition
Oriental	Collectivist	Values the group/community over the individual.
	Holistic	Sees the self as interconnected with others, nature, and the universe.
	Interconnected	Self is deeply linked to society, family, and environment .
Western	Individualistic	Focuses on the self as independent and unique.
	Dualist	Believes the body and soul/mind are separate entities.
	Autonomous	Self is self-governing , can make personal choices without external control.

The Self from a Psychological Perspective	
Psychologist	Theory
Sigmund Freud	Id (pleasure), Ego (reality), Superego (morality).
William James	Self = "I" (subject) and "Me" (object). Me : material, social, spiritual selves.

The Self from a Sociological Perspective	
Thinker	Key Points
George Herbert Mead	Self develops through interaction. "I" = individual, "Me" = socialized self. Generalized and significant others shape self.
Charles Cooley	Looking-glass self : 1) Imagine how we appear 2) Imagine how others judge us 3) Develop self-feelings from judgments.

The Self from an Anthropological Perspective		
Concept	Key Term	Definition
Anthropological Perspective	Holistic Approach	View the self as a whole – includes physical, biological, psychological, social, cultural parts .
	Stratigraphic Self	Layers of the self: Physical, Biological, Psychological, Social .
	Avoid Reductionism	Don't explain everything using just one factor (e.g., biology only).
Cultural Influence on Self	Egocentric	Self-centered identity; values independence (e.g., Americans).
	Sociocentric	Group-centered identity; values harmony with group (e.g., Japanese, Chinese)
Cultural Perspective	Cultural Relativism	Understand people based on their own culture, not yours.
	Ethnocentrism	Belief that your culture is superior to others .
	Xenocentrism	Belief that other cultures are superior to your own .

Western vs Oriental Views of the Self		
View	Key Term	Definition
Oriental	Collectivist	Values the group/community over the individual.
	Holistic	Sees the self as interconnected with others, nature, and the universe.
	Interconnected	Self is deeply linked to society, family, and environment .
Western	Individualistic	Focuses on the self as independent and unique.
	Dualist	Believes the body and soul/mind are separate entities.
	Autonomous	Self is self-governing , can make personal choices without external control.

The Self from a Psychological Perspective	
Psychologist	Theory
Sigmund Freud	Id (pleasure), Ego (reality), Superego (morality).
William James	Self = “ I ” (subject) and “ Me ” (object). Me : material, social, spiritual selves.

Key Term	Definition	Example
Metacognition	Thinking about your own thinking/learning	Alam mong mas natututo ka kapag nagsusulat ka.

The Physical and Sexual Self	
- Adolescence: Early (11–14): Peer-oriented. Middle (15–17): Goal-driven. Late (18–24): Others-focused. - Self-Concept Components: Self-Identity: Who am I? Self-Image: How do I see myself? Self-Concept: Thoughts and feelings about oneself. Self-Esteem: How much do I value myself?	

Sexual Self	
Sex: Biological traits Gender: Cultural identity Sexual Orientation: Pattern of attraction	
Term	Meaning
Cisgender	Gender identity matches birth sex
Transgender	Gender identity differs from birth sex
Intersex	Ambiguous genitalia or reproductive anatomy
Asexual	No sexual attraction
Queer	Non-traditional sexual identity

Unpacking the Self (Other Aspects)	
Material Self: Body + possessions. Spiritual Self: Core identity and meaning. Political Self: National identity, citizenship. Digital Self: Online persona, risks of cyberbullying. Self-Presentation: Managing impressions for internal/external audiences.	

Study Strategy: PQ4R Method		
Key Term	Simple Definition	Example
Preview	Scan the material first	Tinitingnan mo muna ang headings bago magbasa.
Question	Ask questions about the topic	“Ano kaya ang ibig sabihin ng mindset?”
Read	Read the content carefully	Binasa mo ang buong lesson tungkol sa self.
Reflect	Think about what you read	“May koneksyon ba ito sa karanasan ko?”
Recite	Say the ideas in your own words	Ipinaliwanag mo sa kaklase ang natutunan mo.
Review	Go over everything again	Binasa mo ulit bago ang quiz.

Self-Regulation & Internal Motivation		
Key Term	Definition	Example
Self Regulation	Control thoughts, emotions, and actions to reach goals	Hindi ka muna nag-social media para matapos ang homework.
Adherence	Doing something due to rules or pressure	Nag-aaral ka kasi may utos si teacher.
Identification	Doing something because it matches your values	Nag-aaral ka ng science kasi gusto mong maging doktor.
Internalization	Fully accepting behavior as part of self	Araw-araw ka nag-aaral kasi mahalaga ito sa'yo bilang tao.

Motivation & Learning Theories		
Theory	Key Term	Example
Bandura	Self-Efficacy	“Kaya kong gawin ‘to. Nasanay na akong magsalita sa harap.”
Dweck	Fixed Mindset	“Hindi ko kaya ‘yan. Di ako magaling dyan.”
	Growth Mindset	“Mahirap pero matututo rin ako.”
Goal-Setting Theory	5 Elements (Clarity, Challenge, Commitment, Feedback, Task Complexity)	Gusto mo mag-top 5: malinaw ang goal, tinatanggap mo ang challenge, nagpapa-check ka sa teacher.

ETHICS
• Ethics: Study of right and wrong. • Ethos (Greek) = Character • Greek View: Aim is to live a good life. • Judeo-Christian View: Aim is righteousness before God.

Applied Ethics Fields
1. Bioethics – Life & health care issues 2. Business Ethics – Corporate morality 3. Environmental Ethics – Nature and sustainability 4. Sexual Ethics – Human sexual behavior

Moral vs Non-Moral Standards	
Moral Standards	Non-Moral Standards
Cause guilt	Cause shame
Based on reason and universal laws	Based on etiquette, norms

Moral Decision-Making
• Human Act vs Act of Human: 1. Human Act: With intellect and will (moral) 2. Act of Human: Involuntary (e.g., blinking) • 3 Determinants of Moral Act: 1. Object 2. Purpose 3. Circumstances • 7-Step Ethical Decision Model: 1. Gather facts 2. Identify ethical issues 3. Identify principles/virtues 4. List options 5. Compare options 6. Consider consequences 7. Decide

Theories of Ethics	
Theory	Key Ideas
Consequentialism (Teleological)	Morality = good consequences (Bentham, Mill)
Deontology (Kant)	Morality = duty and intent
Natural Law (Aquinas)	Morality = follow reason & nature
Virtue Ethics (Aristotle)	Morality = develop good character
Love & Justice Framework	Morality = promoting common good

Moral Agent & Character		
Key Term	Definition	Example
Character	Set of moral and mental qualities	Marespeto, matapat, mapagkumbaba.
Personality	Your attitude and behavior	Masayahin, tahimik, palabiro.
Moral Agent	The person doing the moral act	Ikaw ‘yung gumawa ng mabuti o masama.
Defining Moments	Big events that shape your values	Pagpili mo ng tama kahit mahirap — nakabago sa paniniwala mo.

Stages of Moral Development (Kohlberg)		
Stage	Focus of Morality	Simple View
Pre-Conventional	Based on reward and punishment	“Di ko gagawin ‘yan kasi baka maparusahan ako.”
Conventional	Based on social approval and rules	“Gagawin ko ‘to kasi ito ang tama sa paningin ng iba.”
Post-Conventional	Based on personal moral principles	“Gagawin ko ‘to dahil naniniwala akong ito ang tama.”

Ethical Frameworks Summary	
Framework	Focus
Virtue Ethics (Aristotle)	Good person = Good actions
Natural Law (Aquinas)	Reason, Conscience, God’s Law
Deontology (Kant)	Duty, not outcome
Utilitarianism (Bentham, Mill)	Greatest good for the greatest number
Justice & Common Good	Equal access and shared resources

Subjectivism & Emotion in Ethics		
Key Term	Definition	Example
Ethical Subjectivism	Moral choices are based on personal approval/disapproval	“Agree ako diyan, so tama ‘yan para sa akin.”
Emotivism	Moral statements express feelings, not facts	“Stealing is bad” = “Boo to stealing!”
Rule of Reason	Moral judgments must be rational , not purely emotional	Di sapat ang feelings lang — kailangan may dahilan.
Feelings	Emotions can help or block good moral decisions	Ang galit minsan nakakasira sa tamang desisyon.

Modes of Cultural Transmission		
Key Term	Simple Definition	Realistic Example
Enculturation	Learning your own culture from birth	A Filipino child grows up learning to say “ po ” and “ opo ”, joins fiestas, and respects elders.
Acculturation	Adapting to a new culture while keeping your original one	A Filipino student studying in Japan learns to bow and follow Japanese etiquette, but still practices Filipino customs at home.
Assimilation	Fully adopting another culture, slowly losing your original one	A Filipino migrant in the U.S. who speaks only English, celebrates only American holidays, and no longer observes Filipino traditions.

ART APPRECIATION

Fundamentals Of Art		
Topic	Key Points	Examples
Definition of Art	- Expression of human creativity - Derived from Latin "ars" (craft)	Carpentry, painting, sculpture
Essential Requirements	- Must be man-made - Beneficial - Uses a medium	Music, visual art
Sources of Beauty	1. Nature – “Mother of all arts” 2. Art – man-made	Mountains, paintings

Philosophy Of Art	
Philosopher	View
Plato	Art is imitation (mimesis)
Aristotle	Art reflects reality and teaches
Kant	Beauty is subjective but can be universal
Tolstoy	Art communicates emotions

ELEMENTS & PRINCIPLES OF ART		
Element	Definition	Types/Examples
Line	Path of a moving point	Zigzag, curved, vertical
Shape	2D enclosed space	Geometric (square), Organic (leaf)
Form	3D structure	Cube, sphere
Color	Reflected light	Primary, Secondary, Tertiary
Value	Lightness/darkness	Shade = add black, Tint = add white
Texture	Surface quality	Physical (real), Implied (illusion)
Space	Area between/around objects	Positive & negative space

KINDS & CLASSIFICATION OF ART		
Type	Description	Examples
Fine/Aesthetic Arts	Meant for admiration	Music, painting, sculpture, architecture, dance
Practical/Useful Arts	Applied to everyday life	Industrial arts, household arts, commercial art
Representational (Objective)	Shows recognizable subjects	Portraits, landscapes
Non-Representational (Non-Objective)	No subject matter	Music, architecture

FUNCTIONS OF ART		
Function	Meaning	Example
1. Aesthetic	Makes us appreciate beauty	A painting of a sunset that makes you feel calm and amazed
2. Utilitarian	Art with a practical use in daily life	A well-designed chair that's both comfortable and stylish
3. Cultural	Preserves traditions and identity	Indigenous weaving that tells a story of a tribe's history
4. Social	Expresses social ideas and promotes unity	A mural that encourages people to help protect the environment

SUBJECTS & METHODS OF ART		
Method	Definition	Examples
Realism	Shows reality as is	Gustave Courbet
Abstraction	Shows artist's feelings	Picasso, Kandinsky
Symbolism	Uses symbols	Spoliarium
Fauvism	Wild colors, spontaneity	Matisse

Dadaism	Nonsense art	Ready-made urinal
Futurism	Celebrates technology	Machines in motion
Surrealism	Dream-like	Salvador Dali
Impressionism	Captures light/moment	Claude Monet

Color Psychology		
Color	Positive Traits	Negative Traits
Red	Strength, energy	Aggression, strain
Blue	Trust, logic	Coldness
Yellow	Confidence	Irrationality
Green	Peace, balance	Boredom
Violet	Luxury, spirituality	Inferiority
Orange	Comfort, fun	Frivolity
Pink	Love, nurture	Weakness
Gray	Neutrality	Depression
Black	Elegance	Oppression
White	Purity, simplicity	Coldness
Brown	Earthiness, reliability	Heaviness

Philippine Art Timeline	
Features	Example Artworks
Prehistoric Art (40,000–4,000 BC)	
Natural pigments, stone carvings, rituals, cave paintings, portable and stationary art	1. Angono Petroglyphs – 127 figural carvings on a volcanic cave wall; oldest known artworks 2. Manunggul Jar – secondary burial jar from Tabon Caves, Palawan 3. Maitum Jar – earthenware burial jars found in Ayub Cave, Sarangani
Ancient Art (4000 BC – 400 AD)	
Civilizations with writing systems , used art to tell stories, decorate tools , show religion, social class , and power	1. Code of Hammurabi – Babylonian stone-carved laws 2. Parthenon – temple for Athena 3. Venus de Milo – sculpture 4. Kerch vases – late Greek red-figure pottery

	5. Krater – Greek wine-mixing vase 6. Pelike – Greek pottery with two vertical handles
Medieval Art (500–1400 AD)	
Known as the “ Dark Ages ”, focused on the Church , art became religious , included biblical scenes, mythology, manuscripts , and Gothic architecture	1. Rose window 2. Mosaic of Jesus Christ – Istanbul, Turkey 3. Hagia Sophia – Constantinople 4. Lindisfarne Gospels – illuminated manuscript 5. Byzantine mosaics – Palatine Chapel, Sicily 6. Notre-Dame Cathedral
Renaissance Art (1400–1600)	
Rebirth of art, science, and culture in Italy Focused on humanism, individuality, nature , and realism	1. Leonardo da Vinci – <i>Mona Lisa, The Virgin of the Rocks, The Last Supper, Vitruvian Man</i> 2. Michelangelo – <i>Pietà, David, Sistine Chapel Ceiling</i> 3. Raphael – <i>Sistine Madonna, Madonna of the Chair, The School of Athens</i> 4. Brunelleschi – <i>Duomo of Florence</i> 5. Donatello – <i>Bronze David</i> 6. Botticelli – <i>Primavera, Adoration of the Magi, Birth of Venus</i> 7. Bellini – <i>Portrait of Doge Loredano</i>
Baroque Art (1600–1750)	
Grand, emotional , and dramatic ; often religious , with strong use of light and shadow	a. Caravaggio – <i>Death of the Virgin</i> b. Bernini – Baroque sculpture pioneer c. Velázquez – <i>Portrait of Philip IV, Las Meninas</i> d. Rubens – <i>Assumption of the Virgin, Judgement of Paris, The Garden of Love</i> e. Rembrandt – <i>The Night Watch, Man with the Golden Helmet, Descent from the Cross</i>
Rococo Art (1700–1800)	
Known for lightness, elegance , and curved ornamentation ; based on fêtes galantes (romantic courtship scenes)	Jean Antoine Watteau – <i>La Surprise</i>
Neoclassicism (Late 1700s – Early 1800s)	
Focused on order, heroism , and classical	Jacques-Louis David – <i>Oath of the Horatii, Death of</i>

ideals; art was serious and moral	<i>Socrates, Death of Marat, Bonaparte Crossing the Alps</i> Jean-Auguste-Dominique Ingres – <i>La Grande Odalisque, Vow of Louis XIII, The Apotheosis of Homer, The Turkish Bath</i>
Romanticism (Late 1700s – Mid 1800s)	
Highlighted emotion, imagination, freedom, and individualism ; reacted against strict rules of Neoclassicism	Francisco Goya – <i>The Black Duchess, The Nude Maja, The Third of May 1808, Saturn Devouring His Son</i> Théodore Géricault – <i>The Raft of the Medusa, Portrait of Mustapha</i> Eugène Delacroix – <i>Scenes from the Massacres of Chios, The Death of Sardanapalus, Liberty Leading the People, Apollo Slaying the Serpent</i>

Asian & Western Art

Asian Art	
Aspect	Details
Key Features	- Rich in history and culture - Known for calligraphy, bronze, jade, temples, textiles, and landscape painting - Calligraphy = highest art in East Asia
Traditional Artists & Works	- Fan Kuan – <i>Travelers Among Mountains and Streams</i> - Hokusai – <i>Thirty-Six Views of Mount Fuji</i> - Basawan – <i>Akbar Restraining the Enraged Elephant</i>
Modern Art in Asia	- Fast-growing contemporary scene - More art festivals, museums, and global exposure
Famous Modern Artists	- Cai Guo-Qiang (China) – gunpowder art - Miwa Yanagi (Japan) – photo & feminist art - Suh Do-Ho (Korea) – installation art - Rirkrit Tiravanija (Thailand) – social & interactive art

Western Art	
Period	Main Idea
Classical	From Ancient Greece & Rome ; focus on beauty, balance, and order
Byzantine	Religious art with gold, icons, and flat images
Medieval	Centered on the Church ; cathedrals, stained glass, manuscripts
Renaissance	"Rebirth" of classical learning; focused on realism, humanity, and science
Neoclassicism	Return to Greek/Roman styles ; showed heroism, discipline, and reason
Romanticism	Highlighted emotion, nature, and freedom
Realism	Showed real life and social issues, no idealism
Impressionism	Captured light, color, and quick moments in daily life
Modernism	Broke tradition; included abstract and experimental styles
Postmodernism	Mixed styles, irony, digital, and conceptual art ; challenged art rules

Music History		
Period	Features	Examples
Medieval	Gregorian chant, organum	Lyra, dulcimer, flute
Ars Nova	Polyphonic development	Philippe de Vitry
Baroque	Ornate, emotional	Bach, Handel, Vivaldi
Classical	Structured, balanced	Mozart, Beethoven

Forms:

- **Sonata** – 3–4 movements
- **Symphony** – Orchestral sonata
- **Concerto** – Solo + orchestra
- **Opera** – Storytelling with music

Beethoven: Transitioned from Classical to Romantic; wrote 9 symphonies, became deaf but continued composing.

National Artists of the Philippines	
Name	Works/Known
Fernando Amorsolo <i>(Painting)</i>	"Grand Old Man of Philippine Art" , <i>Planting Rice, Dalagang Bukid</i>
Guillermo Tolentino <i>(Sculpture)</i>	"Father of Philippine Sculpture" , <i>Oblation, Bonifacio Monument</i>
Carlos "Botong" Francisco <i>(Painting)</i>	<i>The Progress of Medicine</i> , murals in Manila
Napoleon Abueva <i>(Sculpture)</i>	"Father of Modern Philippine Sculpture" , <i>Nine Muses, Kaganapan</i>
Leandro Locsin <i>(Architecture)</i>	<i>CCP Complex, Istana Nurul Iman</i>
Lucio San Pedro <i>(Music)</i>	<i>Sa Ugoy ng Duyan, Lahing Kayumanggi</i>
Levi Celerio <i>(Music)</i>	"Dahon Musician" , <i>Ang Pipit, Sa Ugoy ng Duyan</i>
Francisco Arcellana <i>(Literature)</i>	<i>The Mats</i> , Modern short story pioneer
Nick Joaquin <i>(Literature)</i>	<i>May Day Eve, Portrait of the Artist</i> , "Quijano de Manila"
Amado V. Hernandez <i>(Literature)</i>	"Ama ng Manggagawang Pilipino" , <i>Isang Dipang Langit, Luha ng Buwaya</i>
Bienvenido Lumbera <i>(Literature)</i>	<i>Likhang Dila, Likhang Diwa</i> , Literary scholar and librettist
Virgilio Almario <i>(Literature)</i>	<i>Doktrinang Anakpawis</i> , "Makata ng Bayan" , <i>Mga Retrato at Rekwerdo</i>
Jose Garcia Villa <i>(Literature)</i>	<i>Comma Poems, Footnote to Youth</i> , "Dovegion"
Alejandro Roces <i>(Literature)</i>	<i>My Brother's Peculiar Chicken</i> , Noted humorist
Carlos P. Romulo <i>(Literature)</i>	<i>I Saw the Fall of the Philippines</i> , UN diplomat and journalist
Lamberto Avellana <i>(Film/Theater)</i>	<i>Sakay, Anak Dalita</i> , "Boy Wonder of Philippine Movies"
Daisy Avellana <i>(Theater)</i>	<i>Othello, Casa de Bernarda Alba</i> , First Lady of Philippine Theater

Wilfrido Ma. Guerrero <i>(Theater)</i>	<i>Wanted: A Chaperone, Forever</i>
Salvador Bernal <i>(Theater Design)</i>	Pioneer in stage and costume design , CCP productions
Antonio Molina <i>(Music)</i>	<i>Hatinggabi, Ana Maria</i> , "Claude Debussy of the Philippines"
Andrea Veneracion <i>(Music)</i>	Founder of Philippine Madrigal Singers
Lucrecia Kasilag <i>(Music)</i>	<i>Toccata for Percussion and Winds</i> , "First Lady of Philippine Music"
Ryan Cayabyab <i>(Music)</i>	<i>Kay Ganda ng Ating Musika, Da Coconut Nut</i> , "Mr. C"
Ramon Valera <i>(Fashion Design)</i>	Modernized the terno , First Nat'l Artist for Fashion
Francisca Reyes Aquino <i>(Dance)</i>	<i>Philippine Folk Dances</i> , "Mother of Philippine Folk Dance"
Leonor Orosa-Goquingco <i>(Dance)</i>	<i>Filipinescas</i> , "Mother of Philippine Theater Dance"
Alice Reyes <i>(Dance)</i>	<i>Amada, Itim Asu</i> , Founder of Ballet Philippines
Agnes Locsin <i>(Dance)</i>	<i>Encantada</i> , Neo-ethnic choreography
Ishmael Bernal <i>(Film)</i>	<i>Himala, Pagdating sa Dulo</i> , Social realist filmmaker
Lino Brocka <i>(Film)</i>	<i>Maynila sa Kuko ng Liwanag, Tinimbang Ka Ngunit Kulang</i>
Kidlat Tahimik <i>(Film)</i>	<i>Perfumed Nightmare</i> , "Father of Indie Philippine Cinema"
Cesar Legaspi <i>(Painting)</i>	<i>Gadgets, Man and Woman</i> , Neo-realist style
Hernando Ocampo <i>(Painting)</i>	<i>Genesis, Calvary</i> , Abstract modernist
Ang Kiukok <i>(Painting)</i>	<i>Fishermen, Crucifixion</i> , Cubist-expressionist painter
Benedicto "BenCab" Cabrera <i>(Painting)</i>	<i>Sabel Series</i> , "Master of Contemporary Philippine Art"
Jose Joya <i>(Painting)</i>	<i>Space Transfiguration, Granadean Arabesque</i> , Abstract expressionism
Arturo Luz <i>(Painting)</i>	<i>Cyclist, Bagong Taon</i> , Minimalist style

Vicente Manansala <i>(Painting)</i>	<i>Jeepneys, Madonna of the Slums, Transparent Cubism</i>
Federico Alcuaz <i>(Painting)</i>	<i>Barcelona Series, Aubusson Tapestries</i>
J. Elizalde Navarro <i>(Mixed Media)</i>	<i>Maskara Series, known for mixed media art</i>
Juan Nakpil <i>(Architecture)</i>	<i>Quiapo Church renovation, First National Artist for Architecture</i>
Pablo Antonio <i>(Architecture)</i>	<i>FEU campus, Ideal Theater, Art Deco style</i>
Jose Ma. Zaragoza <i>(Architecture)</i>	<i>Santo Domingo Church, Meralco Building</i>
Jose Maceda <i>(Music)</i>	<i>Udlot-Udlot, Pagsamba, Ethnomusicologist</i>
Francisco Feliciano <i>(Music)</i>	<i>La Loba Negra, Pokpok Alimpako</i>
Manuel Conde <i>(Film)</i>	<i>Genghis Khan, Juan Tamad Series</i>
Antonio Buenaventura <i>(Music)</i>	<i>Pandanggo sa Ilaw, Echoes of the Past</i>
Carlos Quirino <i>(Historical Literature)</i>	<i>The Great Malayan (Rizal), Only Nat'l Artist for Historical Lit</i>
Julio Nakpil <i>(Music)</i>	<i>Salve Patria, Pasig Panting, Composer of the Revolution</i>
Francisca Reyes Aquino <i>(Dance)</i>	<i>"Mother of Philippine Folk Dance", Philippine Folk Dances Vol. I–V</i>
Leonor Orosa-Goquingco <i>(Dance)</i>	<i>"Mother of Philippine Theater Dance", Filipinescas: Philippine Life, Legend and Lore in Dance</i>
Ramon Obusan <i>(Dance)</i>	<i>Cultural researcher & choreographer, Vamos a Belen, Noon Po sa Amin</i>
Alice Reyes <i>(Dance)</i>	<i>Founder of Ballet Philippines, Amada, Itim Asu</i>
Agnes Locsin <i>(Dance)</i>	<i>Neo-ethnic choreographer, Encantada, Sayaw, Sabel</i>
Juan F. Nakpil <i>(Architecture)</i>	<i>"First National Artist for Architecture", designer of Quezon Hall (UP Diliman), Rizal Shrine Restoration</i>
Pablo S. Antonio <i>(Architecture)</i>	<i>"Pioneer of Modern Philippine Architecture", designer of FEU buildings, Ideal Theater</i>

Leandro V. Locsin <i>(Architecture)</i>	"Architect of the CCP" , designer of <i>PICC, Istana Nurul Iman (Brunei)</i>
Ildefonso P. Santos Jr. <i>(Landscape Architecture)</i>	"Father of Philippine Landscape Architecture" , known for <i>Paco Park, Luneta Park</i>
José María Zaragoza <i>(Architecture)</i>	Innovator in Modern Church Architecture , designed <i>Santo Domingo Church, Meritxell Chapel (Spain)</i>
Francisco “Bobby” Mañosa <i>(Architecture)</i>	"Champion of Filipino Architecture" , famous for <i>Coconut Palace, EDSA Shrine, use of bahay kubo vernacular designs</i>

Famous Personalities in Art		
Artist	Country	Famous Works
Leonardo da Vinci	Italy	<i>The Last Supper, Mona Lisa</i>
Vincent van Gogh	Netherlands	<i>Starry Night, Sunflowers</i>
Michelangelo	Italy	<i>David, Creation of Adam</i>
Claude Monet	France	<i>Water Lilies, Impression, Sunrise</i>
Frida Kahlo	Mexico	<i>The Two Fridas, Broken Column</i>
Georgia O’Keeffe	USA	<i>Black Irises, Radiator Building</i>
Edgar Degas	France	<i>The Absinthe Drinker, The Dance Class</i>
Edvard Munch	Norway	<i>The Scream, Dance of Life</i>
Jackson Pollock	USA	<i>Number 5, Number 31</i>
Pablo Picasso	Spain	<i>Guernica, Weeping Woman</i>
Gustav Klimt	Austria	<i>The Kiss, The Virgin</i>
Henry Moore	UK	<i>Reclining Figures, King and Queen</i>

Salvador Dali	Spain	<i>Persistence of Memory</i>
Paul Cézanne	France	<i>The Bathers, The Card Players</i>
Diego Rivera	Mexico	<i>Man at the Crossroads, History of Mexico</i>

SCIENCE

Branch of Science	Study of...
Actinobiology	Study of effects of radiation on living organisms
Agrostology	Study of grasses
Algology (Phycology)	Study of algae
Anthozoology	Study of interaction between humans and animals
Acarology	Study of ticks and mites
Araneology	Study of spiders
Astacology	Study of crayfish
Axiology	Study of values and value judgments (philosophy-science overlap)
Bromatology	Study of food composition and nutrition
Bryology	Study of mosses and liverworts
Carcinology	Study of crustaceans (crabs, lobsters, etc.)
Cetology	Study of whales, dolphins, and porpoises
Chondrology	Study of cartilage
Cladistics	Study of evolutionary relationships between species
Cnidariology	Study of jellyfish and related organisms (Cnidarians)
Conchology	Study of shells and mollusks
Cryobiology	Study of life under low temperatures

Dendrology	Study of trees and shrubs
Enzymology	Study of enzymes
Epidemiology	Study of disease distribution in populations
Euthenics	Study of improving human functioning through environment
Exobiology	Study of life in outer space
Felidology	Study of cats (wild and domestic)
Geobotany	Study of geographic distribution of plants
Geoponics	Study of agriculture
Glyptology	Study of carvings and engraving
Heliology	Study of the sun
Helminthology	Study of parasitic worms
Hepatology	Study of the liver
Hippology	Study of horses
Horology	Study of time and timekeeping
Ichnology	Study of fossilized footprints and traces
Lepidopterology	Study of butterflies and moths
Lichenology	Study of lichens
Malacology	Study of mollusks (snails, clams, octopuses)
Myrmecology	Study of ants
Nanotechnology	Study of manipulating matter at the atomic level
Necrobiology	Study of organisms after death
Neonatology	Study of newborn infants
Nosology	Study of classification of diseases
Oology	Study of eggs (especially bird eggs)

Paleobotany	Study of ancient plants (fossilized)
Palynology	Study of pollen and spores
Pedology	Study of soils
Pomology	Study of fruits and their cultivation
Proctology	Study of the rectum and anus
Psammology	Study of sand
Serology	Study of blood serum and immune responses
Speleology	Study of caves
Tegumentology	Study of body coverings (skin, scales, etc.)
Thanatology	Study of death and practices associated with it

Learning & Developmental Disorders	
Disorder	Short Meaning
Dyslexia	Difficulty reading words.
Dyscalculia	Trouble with math.
Dysgraphia	Poor writing and spelling.
Dyspraxia	Poor coordination or movement.
ADHD (<i>Attention Deficit Hyperactivity Disorder</i>)	Inattentive and hyperactive.
ASD (<i>Autism Spectrum Disorder</i>)	Trouble with social and communication.
Dysnomia	Trouble remembering words.
Speech Delay	Slow language development.
APD (<i>Auditory Processing Disorder</i>)	Trouble understanding sounds.
NVLD (<i>Nonverbal Learning Disorder</i>)	Poor visual or motor skills.

Neurodevelopmental & Mental Disorders	
Disorder	Meaning
Alzheimer’s Disease	Memory loss disease.
Parkinson’s Disease	Movement disorder.
Tourette Syndrome	Involuntary tics or sounds.
OCD (<i>Obsessive-Compulsive Disorder</i>)	Repetitive thoughts and actions.
Anxiety Disorder	Excessive worry.
Depression (<i>MDD - Major Depressive Disorder</i>)	Persistent sadness, no energy.
Bipolar Disorder	Mood swings.
Schizophrenia	Hallucinations and disordered thinking.
PTSD (<i>Post-Traumatic Stress Disorder</i>)	Trauma-related fear and flashbacks.
Anorexia Nervosa (<i>AN</i>)	Starving due to body image issues.

Sensory Disorders (Vision & Hearing)	
Disorder	Meaning
Myopia	Can’t see far.
Hyperopia	Can’t see near.
Astigmatism	Blurry vision.
Presbyopia	Aging eyes, can’t focus near.
Color Blindness	Can’t see some colors.
Cataract	Cloudy lens.
Glaucoma	Eye pressure damages optic nerve.
AMD (<i>Age related Macular Degeneration</i>)	Central vision loss.
Tinnitus	Ringing in ears.
Hearing Loss	Can’t hear well.

Genetic/Chromosomal Disorders	
Disorder	Meaning
Down Syndrome (Trisomy 21)	Extra chromosome 21.
Turner Syndrome	Missing X chromosome (females).
Klinefelter Syndrome (KS)	Extra X chromosome in males (XXY).
Cystic Fibrosis (CF)	Thick mucus in lungs/digestive tract.
Tay-Sachs Disease	Nerve cell damage.
Hemophilia	Can't stop bleeding.
PKU (Phenylketonuria)	Can't break down protein.
SCA (Sickle Cell Anemia)	Misshaped red blood cells.
Huntington's Disease (HD)	Brain cell degeneration.
Fragile X Syndrome (FXS)	Learning delay (mostly males).

Physical / Motor / Digestive Disorders	
Disorder	Meaning
Cerebral Palsy (CP)	Poor muscle control and movement.
Spina Bifida	Spine doesn't form properly.
Arthritis	Joint swelling and pain.
Osteoporosis	Weak, brittle bones.
GERD (Gastroesophageal Reflux Disease)	Stomach acid backs into throat.
Peptic Ulcer Disease (PUD)	Stomach or intestinal sores.
Lactose Intolerance	Can't digest milk.
IBS (Irritable Bowel Syndrome)	Belly pain with diarrhea or constipation.
Constipation	Hard, dry stools; difficult bowel.
Obesity	Excess body fat.

Earth And Space Phenomena
Solar Eclipse - Sun is blocked by the Moon. Alignment: Sun – Moon – Earth (SME). - The Moon blocks the Sun's light from reaching Earth. - Caused by the Moon coming to light and casting a shadow on Earth. Lunar Eclipse - Earth is blocked by the Moon. Alignment: Sun – Earth – Moon (SEM). - Happens at Nighttime. - The Earth casts a shadow on the Moon. - Caused by the Earth's shadow between the Sun and the Moon, casting a shadow on the Moon.

Planets in the Solar System		
Terrestrial Planets (Rocky Planets or Telluric Planets)		
Planet	Description	Trivia
Mercury	Smallest planet; no moons; closest to the Sun.	A year on Mercury is only 88 Earth days long. It has extreme temperatures: -173°C at night, 427°C by day .
Venus	Hottest planet; no moons; thick CO ₂ atmosphere traps heat via the greenhouse effect .	Venus rotates backward (retrograde rotation) and has the slowest rotation - a day is longer than its year!
Earth	Only known planet with liquid water and life; has one moon.	Earth's moon helps stabilize its axial tilt, affecting seasons . Earth is the densest planet in the solar system.
Mars	Known as the Red Planet due to iron oxide (rust) on its surface; has 2 moons: Phobos & Deimos .	Mars has the largest volcano (Olympus Mons) and deepest canyon (Valles Marineris) in the solar system.
Jovian Planets (Giant Planets or Gas Planets)		
Jupiter	Largest planet; gas giant; has a strong magnetic field and over 90 moons .	The Great Red Spot is a massive storm larger than Earth that has lasted for centuries .

Saturn	Gas giant with spectacular ring system ; has the most known moons (over 150 including candidates).	Saturn's moon Titan is larger than Mercury and has a thick atmosphere with liquid methane lakes .
Uranus	Ice giant; bluish due to methane; coldest planet; rotates sideways (axis tilt ~98°) .	Discovered by William Herschel in 1781; first planet discovered with a telescope.
Neptune	Ice giant; farthest from the Sun; dark blue color from methane.	Winds on Neptune can reach up to 2,100 km/h , the fastest recorded in the solar system.

Earth and Celestial Phenomena:	
Phenomenon	Cause / Description
Tides	Caused by the gravitational pull of the Moon (main influence) and the Sun on Earth's oceans. Results in high and low tides .
Day and Night	Caused by Earth's rotation on its axis every 24 hours . One side of Earth faces the Sun (day), the other side is in shadow (night).
Seasons	- Caused by Earth's revolution around the Sun (~365.25 days). - Due to Earth's tilted axis (23.5°), which affects sunlight distribution in each hemisphere.
Solstices & Equinoxes	Caused by the tilt of Earth's axis and its orbit around the Sun .
Summer Solstice	Occurs around June 21 (Northern Hemisphere); longest day of the year when the Sun is directly overhead at the Tropic of Cancer .
Winter Solstice	Occurs around December 21 (Northern Hemisphere); shortest day of the year when the Sun is directly over the Tropic of Capricorn .

Autumnal Equinox	Occurs around September 23 ; equal length of day and night (12 hours). The Sun is directly over the equator .
Vernal Equinox	Occurs around March 21 ; also equal day and night.
Midnight Sun	Occurs in polar regions (e.g., Arctic Circle) during summer, where the Sun is visible for 24 hours due to the Earth's tilt.
Moon	Earth's natural satellite ; orbits Earth every 27.3 days . It reflects sunlight (does not emit its own light). Affects tides and lunar phases.
Perigee	The Moon is at its closest point to Earth (~363,300 km); appears slightly larger (often called a supermoon).
Apogee	The Moon is at its farthest point from Earth (~405,500 km); appears smaller (often called a micromoon).
Sun	A G-type main-sequence star ; contains 99.8% of the solar system's mass . Light takes 8 minutes to reach Earth, but thousands of years to escape the core.
Perihelion	Earth is at its closest point to the Sun (~147 million km); occurs around early January .
Aphelion	Earth is at its farthest point from the Sun (~152 million km); occurs around early July .

Mohs scale

Mohs scale measures mineral hardness by how easily one mineral scratches another, from 1 (softest) to 10 (hardest).

Mineral	Description / Example
Talc	Softest mineral
Gypsum	Can be scratched by a fingernail
Calcite	Can be scratched by a copper coin
Fluorite	Can be scratched by a steel knife

Apatite	Can be scratched by a knife
Orthoclase (Feldspar)	Common in granite; can scratch glass
Quartz	Can scratch steel and glass
Topaz	Harder than quartz
Corundum	Includes sapphire and ruby
Diamond	Hardest mineral

Layers of the Atmosphere (TrosMITE)		
Layer	Description	Trivia
Troposphere	- Lowest layer. - Where weather, clouds, and life-supporting air exist. - Altitude: 0–12 km.	Contains about 75% of the atmosphere's mass. Temperature decreases with altitude in this layer.
Stratosphere	- Contains the ozone layer , which absorbs harmful UV rays. - Altitude: 12–50 km.	Commercial jets fly here to avoid turbulence. Temperature increases with altitude due to ozone absorption of sunlight.
Mesosphere	- Coldest layer. - Meteors burn up here. - Altitude: 50–85 km.	Hardest layer to study —too high for planes, too low for satellites. Temperatures can drop to -90°C .
Thermosphere	- Very hot layer (up to 2,000°C), but feels cold due to low particle density. - Where auroras and space shuttles are found. - Altitude: 85–600 km.	Despite high temperature, you'd feel cold here because particles are too sparse to transfer heat.
Ionosphere	- Overlaps Mesosphere and Thermosphere. - Ionized by solar radiation. - Crucial for radio and GPS signals .	Auroras (Northern and Southern Lights) occur here when solar particles collide with gas particles.

Exosphere	- Outermost layer ; merges with outer space. - Where satellites orbit . - Altitude: 600–10,000 km.	Air is extremely thin—atoms can escape into space. This is technically where space begins .
-----------	--	--

Additional Notes:

- Earth’s **atmosphere** helps **regulate temperature, support life, and protect from meteors and radiation.**
- The term “**gas giant**” applies to Jupiter and Saturn, while **Uranus and Neptune** are **ice giants** due to heavier elements like water, methane, and ammonia.
- The **boundary between Earth and space** is known as the **Kármán Line** (~100 km above sea level).

Key Biological Components

- **Capsid** - protein shell of a virus.
- **Steroid hormone** - lipids.
- **Gravity** - The force that holds a configuration of three celestial bodies in a gravitational system.
- **Platelets** - Irregularly shaped, colorless blood cells. Its main function is to help stop bleeding and to help blood clots.
- **Plasma** - Yellowish liquid component of blood, where elements are suspended.
- **Lymph** - Fluid containing white blood cells (lymphocytes), which can help fight infection and carry waste from cells.
- **Red blood cells** - Red blood cells produce the plasma that helps carry oxygen from your blood vessels.
- **White blood cells** - White blood cells fight infections and other diseases.
- **Hemoglobin** - Protein in red blood cells that carries oxygen from your body.
- **Myoglobin** - Myoglobin also carries carbon dioxide from your body.

Nonmetal Properties (DI-PWS-FS)

1. **Non-magnetic** - Not attracted to a magnetic field.
2. **Dull appearance** - Gold, Silver, Bismuth, Copper, etc.
3. **Paramagnetic** - Materials **weakly** attracted to a magnetic field.
4. **Ductile/Malleable** - Can be drawn into wires.
5. **Brittle** - Materials that break easily.
6. **Strongly magnetic** - Materials that are strongly attracted by magnets.

Scientific Instruments - Measurements

1. **Thermometer** - Measures temperature.
2. **Barometer** - Measures air pressure.
3. **Sphygmomanometer** - Measures blood pressure.

4. **Stethoscope** - Measures sounds in the heart, lungs, and blood flow.
5. **Microscope** - Used to see tiny objects.
6. **Telescope** - Used to observe distant objects in space.
7. **Glucometer** - Measures blood sugar levels.

Earthquake Types

1. **Focus (Hypocenter)** - The origin of an earthquake inside the Earth's crust.
2. **Epicenter** - The point on Earth's surface directly above the focus.
3. **Magnitude** - Measures the energy released by an earthquake.
4. **Intensity** - Measures the degree of shaking caused by an earthquake.

Types of Lenses (Vision Correction, etc.)

1. **Convex Lens (Converging Lens)** - Used to correct Farsightedness (Hyperopia).
 - **Shape** - Thicker in the middle, thinner at the edges.
 - **Examples** - Magnifying glass, eyeglasses for farsightedness, camera lenses.
 - **Light** - Light converges to a single curved outward.
2. **Concave Lens (Diverging Lens)** - Used to treat Nearsightedness (Myopia).
 - **Shape** - Thinner in the middle, thicker at the edges.
 - **Examples** - Flashlights, peepholes.
 - **Light** - Light scatters.

Endocrine System & Hormones

1. **Pituitary Gland** - Master gland, vital in regulation, growth & metabolism.
2. **Thyroid Gland** - Located at the base of the neck.
 - Produces hormones that regulate metabolism, heart rate, and body temperature.
 - Hormones produced by the thyroid include T3 & T4.

Changing State Of Matter

- **Melting** - Changing from solid to liquid (e.g., ice to water).
- **Freezing** - Changing from liquid to solid (e.g., water to ice).
- **Evaporation** - Changing from liquid to gas (e.g., water to vapor).
- **Condensation** - Changing from gas to liquid (e.g., water vapor to dew).
- **Sublimation** - Changing from solid to gas (e.g., dry ice to carbon dioxide gas).
- **Deposition** - Changing from gas to solid (e.g., water vapor to frost).

Water Cycle (Steps)

- **Evaporation** - Liquid to gas (water to water air).
- **Transpiration** - Liquid to gas (plants to water air).
- **Condensation** - Water vapor to liquid in clouds.
- **Precipitation** - Water droplets falling from clouds as rain, snow, hail, or sleet.
- **Runoff** - Water flowing over land into rivers, lakes, and oceans.
- **Infiltration** - Liquid water sinking into the ground.

Earthquakes

1. **Aesthenosphere** - A partially molten layer in the upper mantle of the Earth, where plates move.
2. **Seismograph** - Measures the strength and duration of earthquakes.

Human Body & Nervous System

1. **Light Reflexes** -
 - **Pupillary Reflex** - Contraction of pupils in response to light.
 - **Accommodation Reflex** - Focusing on nearby objects.
 - **Corneal Reflex** - Blinking in response to touching the cornea.
 - **Withdrawal Reflex** - Pulling away from a harmful stimulus.
2. **Brain** - Central processing unit of the body.
 - **Cerebrum** - Voluntary movements, sensory processing, language, memory.
 - **Cerebellum** - Balance, posture, coordination.
 - **Brainstem** - Controls automatic body functions like breathing and heart rate.
 - **Limbic System** - Emotions, memory, and motivation.
3. **Lobes of the Brain** -
 - **Frontal Lobe** - Executive functions, decision-making, planning.
 - **Parietal Lobe** - Sensory information, spatial awareness.
 - **Temporal Lobe** - Auditory processing, memory.
 - **Occipital Lobe** - Visual processing.

Hemophilia

1. Genetic mutation that affects blood clotting.
2. Excessive bleeding.

Storm Names by PAGASA

1. **Atlantic Ocean** - Hurricane.
2. **Pacific Ocean** - Typhoon.
Indian Ocean - Cyclone.

PAGASA—PH Atmospheric, Geophysical, and Astronomical Services Administration)

Capacity of the Atmosphere

1. **Air Temperature** - Tropical and warm are able to hold more.
2. **Air Pressure** - Warm and humid throughout the year.
3. **Seasonality** - Warm and humid throughout the year.

Capacity of the Atmosphere

1. **TSWS No. 1** - Minimal (30-60 kph).
2. **TSWS No. 2** - Moderate (61-120 kph).
3. **TSWS No. 3** - Strong (121-170 kph).
4. **TSWS No. 4** - Very Strong (171-220 kph).
5. **TSWS No. 5** - Super Typhoon (221 kph and above).

Tropical Cyclone Formation

1. **Warm ocean waters** - Above 26.5°C (80°F).
2. **Low wind shear** - Minimal change in wind speed or direction.
3. **High humidity** - Abundant moisture.
4. **Low pressure area** - Leads to rising air.

Heat Transfer

1. **Conduction** - Direct contact.
 - Heat transfer without motion.
2. **Convection** - Transfer of heat by movement of fluid.
3. **Radiation** - Transfer of heat by electromagnetic waves.
 - Example: Sunlight heating homes.

Composition of Disaccharides

- **Maltose** - Glucose + Glucose
- **Lactose** - Glucose + Galactose
- **Sucrose** - Glucose + Fructose

Fathers of Various Fields

Biology & Life Sciences

- **Father of Modern Biology / Father of Ancient Taxonomy** - Aristotle
- **Father of Modern Taxonomy / Father of Classification** - Carolus Linnaeus
- **Father of Genetics** - Gregor Mendel
- **Father of Medicine** - Hippocrates
- **Father of Microbiology** - Louis Pasteur
- **Father of Viruses** - Dmitri Ivanovsky
- **Father of Immunology** - Edward Jenner
- **Father of Biochemistry** - Carl Alexander Neuberg
- **Father of Zoology** - Aristotle

Psychology & Behavior

- **Father of Modern Psychology** - Wilhelm Wundt
- **Father of Behaviorism** - John B. Watson
- **Father of Psychoanalysis** - Sigmund Freud
- **Father of Psychophysics** - Gustav Fechner
- **Father of Cognitive Psychology** - Jean Piaget

Science & Mathematics

- **Father of Modern Science / Telescope** - Galileo Galilei
- **Father of the Periodic Table** - Dmitri Mendeleev
- **Father of Computer Science** - Alan Turing
- **Father of Linear Programming** - George Dantzig
- **Father of History** - Herodotus
- **Father of Economics** - Adam Smith
- **Father of Modern Management** - Peter Drucker
- **Father of Modern Philosophy** - René Descartes
- **Father of Modern Physics** - Albert Einstein
- **Father of Chemistry** - Antoine Lavoisier
- **Father of Modern Mathematics** - Isaac Newton
- **Father of Astronomy** - Nicolaus Copernicus
- **Father of Geometry** - Euclid

Other Contributions

- **Theory of Evolution / Natural Selection** - Charles Darwin
- **Theory of Drills** - Frederick Winslow Taylor

Etymology/Classification

- **Domain** - Eukarya
- **Kingdom** - Animalia
- **Phylum** - Chordata
- **Class** - Mammalia
- **Order** - Primates
- **Family** - Hominidae
- **Genus** - Homo
- **Species** - Sapiens

New Taxonomy of Life

1. **Animals** - Fauna (Heterotrophic Multicellular)
2. **Plants** - Flora (Autotrophic Multicellular)
3. **Fungi** - Fungi (Decomposers)
4. **Protists** - Single-celled Eukaryotes, diverse.
5. **Monera** - Unicellular Prokaryotes, diverse.

Key Biological Concepts

- **Autotrophs** - Organisms that produce their own food using sunlight or chemicals.
- **Heterotrophs** - Organisms that obtain food by consuming other organisms.
- **Prokaryotes** - Simple cells, no nucleus, no membrane-bound organelles (e.g., bacteria).
- **Eukaryotes** - Complex cells, nucleus, membrane-bound organelles (e.g., plants, animals, fungi, protists).

2 Classifications of Cells (PUE-M)

1. **Prokaryotic** - Single-celled organisms with no true nucleus.
 - Example: Bacteria
2. **Eukaryotic** - Organisms with a true nucleus.
 - Example: Fungi, Animals, Plants, Protists

Asexual Reproductions (F-F-B-B-G-S-R-F-F-S)

1. **Fission** - Asexual reproduction in which an organism splits into two or more individuals.
 - Example: Bacteria
2. **Fragmentation** - Asexual reproduction where a whole new individual grows from a fragment of the parent.
 - Example: Starfish
3. **Budding** - Asexual reproduction where a new organism develops from an outgrowth or bud due to cell division at one particular site.
 - Example: Yeast
4. **Binary Fission** - Asexual reproduction where a cell divides into two identical daughter cells.
 - Example: Amoeba
5. **Gemmules** - Internal buds formed by sponges for asexual reproduction.
6. **Spores** - Reproductive cells produced by fungi, plants, and algae for asexual reproduction.
7. **Regeneration** - The process of regrowing lost body parts or forming a whole new organism from a fragment.
 - Example: Flatworms
8. **Vegetative Propagation** - Asexual reproduction in plants using vegetative parts like stems, leaves, or roots.
 - Example: Potatoes
9. **Fission (Plants)** - A form of asexual reproduction in some plants where the plant divides into two or more parts, each capable of developing into a new plant.

Plant Hormones

- **Auxins** - Responsible for fruit ripening and root development.
- **Cytokinins** - Regulate cell division and cell differentiation.
- **Gibberellins** - Promote stem elongation, seed germination, and flowering.
- **Abscisic Acid** - Inhibits growth, promotes dormancy, and helps plants respond to stress.
- **Ethylene** - Promotes fruit ripening and senescence.

Animal vs. Plant Cell Differences

- **Cell Walls** - Present in plant cells, absent in animal cells. Provide structural support.
- **Chloroplasts** - Present in plant cells, absent in animal cells. Site of photosynthesis.
- **Vacuoles** - Larger in plant cells, smaller or absent in animal cells. Store water, nutrients, and waste.
- **Centrioles** - Present in animal cells, absent in plant cells. Involved in cell division.

Cell Diagram - Basic Unit of Structure and Function in Living Things

- **Nucleus** - Controls cell activities, contains genetic material.
- **Cell Membrane** - Controls what enters and exits the cell.

- **Cytoplasm** - Jelly-like substance that fills the cell, where organelles are suspended.
- **Mitochondria** - Powerhouse of the cell, produces energy (ATP).
- **Ribosomes** - Site of protein synthesis.
- **Endoplasmic Reticulum** - Network of membranes involved in protein and lipid synthesis.
- **Golgi Apparatus** - Modifies, sorts, and packages proteins and lipids.
- **Lysosomes** - Break down waste materials and cellular debris.
- **Vacuole** - Storage sac for water, nutrients, and waste.
- **Cell Wall** - Rigid outer layer in plant cells, provides support and protection.
- **Chloroplast** - Site of photosynthesis in plant cells.

Most Abundant Compounds & Elements (HWOON)

- **Water** - Most abundant inorganic compound.
- **Carbon** - Most abundant organic element.
- **Hydrogen** - Most abundant element in the universe.
- **Oxygen** - Most abundant element in the Earth's crust and atmosphere.
- **Nitrogen** - Most abundant element in Earth's atmosphere.

Joints & Movement - (C-P-S-M-P)

- **Ball-and-Socket Joint** - Allows movement in all directions (e.g., shoulder, hip).
- **Hinge Joint** - Allows movement in one plane (e.g., knee, elbow).
- **Pivot Joint** - Allows rotation (e.g., neck, forearm).
- **Gliding Joint** - Allows limited sliding movement (e.g., wrist, ankle).
- **Saddle Joint** - Allows movement in two planes (e.g., thumb).

Human Anatomy (Quick Facts)

- **Bones in Human Body** - 206
- **Muscles** - 650
- **Heart Chambers** - 4
- **Brain Weight** - 3 lbs
- **Blood Volume** - 5 liters (1.3 gallons)
- **Spine Vertebrae** - 33
- **Ribs** - 12 pairs
- **Skull Bones** - 22
- **Teeth** - 32 (adult)
- **Smallest Bone in Human Body** - Stapes (in ear)

Basic Biological Concepts

- **Atom** - The smallest unit of matter.
- **Molecule** - Two or more atoms bonded together.
- **Cell** - The basic unit of life.
- **Tissue** - Group of similar cells working together.
- **Organ** - Group of tissues working together.

- **Organ System** - Group of organs working together.
- **Organism** - A living being.
- **Population** - Group of individuals of the same species.
- **Community** - Different populations living together.
- **Ecosystem** - Community and its non-living environment.
- **Biosphere** - All ecosystems on Earth.

Types of Seed Plants

1. **Gymnosperms** - Non-flowering plants (e.g., pines, firs, spruces).
2. **Angiosperms** - Flowering plants (e.g., apple, mango, orange).

Parts of a Flower (P-S-C-O-S-M-A-P)

1. **Pistil** - Female reproductive structure.
2. **Stamen** - Male reproductive structure.
3. **Petals** - Attract pollinators.
4. **Sepals** - Protect the bud.
5. **Ovules** - Contain eggs.
6. **Anther** - Produces pollen.
7. **Filament** - Supports the anther.

Features of Monocots vs. Dicots		
Feature	Monocots	Dicots
Seed	One cotyledon	Two cotyledons
Leaf Shape	Narrow, parallel veins	Broad, net-like veins
Example	Rice, corn, wheat, onion, orchids	Mango, beans, roses, oak

Plant Structures and Functions

1. **Roots** - Anchor the plant, absorb water and nutrients.
2. **Stem** - Supports the plant, transports water and nutrients.
3. **Leaves** - Site of photosynthesis.
4. **Flower** - Reproductive organ of plants.
5. **Fruit** - Protects seeds.
6. **Seed** - Contains embryo of a new plant.
7. **Xylem** - Transports water.
8. **Phloem** - Transports food.
9. **Chlorophyll** - Green pigment in plants, absorbs sunlight.
10. **Photosynthesis** - Process by which plants convert sunlight into energy.
11. **Respiration** - Process by which plants release energy.
12. **Glucose and Oxygen** - Products of photosynthesis.

Body Systems

- **Circulatory System** - Transports blood, oxygen, nutrients, hormones, and waste.
- **Respiratory System** - Exchange of gases.
- **Digestive System** - Breaks down food.
- **Excretory System** - Removes waste.
- **Nervous System** - Controls and coordinates body functions.
- **Skeletal System** - Supports the body.
- **Muscular System** - Movement.
- **Endocrine System** - Produces hormones.
- **Immune System** - Protects against disease.
- **Integumentary System** - Skin, hair, nails.

Components of Blood

1. **Red Blood Cells** - Transport oxygen.
2. **White Blood Cells** - Fight infection.
3. **Platelets** - Help blood clot.
4. **Plasma** - Liquid part of blood.

SYMBIOSIS: The art of living together

1. **Mutualism** - Both organisms benefit (e.g., Bee & Flower).
2. **Commensalism** - One organism benefits, the other is unaffected (e.g., Cattle Egret & Livestock).
3. **Parasitism** - One organism benefits, the other is harmed (e.g., Tick & Dog).
4. **Competition** - Both organisms are harmed (e.g., Lion & Hyena).
5. **Predation** - One organism benefits, the other is killed (e.g., Lion & Zebra).
6. **Ammensalism** - One organism is harmed, the other is unaffected (e.g., Algae Bloom & Fish).

DNA & Chromosomes

1. **Chromosome** - Thread-like structures in the nucleus of cells, carrying genetic information.
 - **X Chromosome** - Female (XX).
 - **Y Chromosome** - Male (XY).
2. **Chromosome Count in Humans** - 46 chromosomes (23 pairs).
3. **Down Syndrome** - Trisomy 21 (caused by an extra 21st chromosome).

Biomes & Biodiversity

- Biomes** - Large regions characterized by their vegetation, soil, climate, and wildlife.
- **Tropical Rainforests** - The largest forests in the world.
 - **Temperate Deciduous Forests** - A type of forest biome.
 - **Coniferous Forests (Taiga)** - Largest land biome.
 - **Grasslands** - Dominated by grasses.
 - ✓ **Prairies** - Grasslands with tall grasses, found in central North America.
 - ✓ **Steppe** - Grasslands with short grasses.
 - ✓ **Savannas** - Grasslands with scattered trees, found in tropical regions.

- **Deserts** - Extremely dry regions with very little precipitation.
 - ✓ **Sahara** - The largest hot desert in the world.
 - ✓ **Antarctica** - The coldest desert in the world.
 - ✓ **Atacama** - The driest desert in the world.
 - ✓ **Sonoran** - The hottest desert in North America.
- **Aquatic Biomes** - Freshwater and saltwater.

Biodiversity - Refers to all the different kinds of life found in an area.

Behavioral Adaptations

1. **Migration** - Animals move from one place to another in search of food or to avoid harsh weather.
2. **Hibernation** - Animals go into a deep sleep during winter to conserve energy.
3. **Estivation** - Animals go into a state of dormancy during hot, dry periods.
4. **Camouflage** - Animals blend in with their surroundings to avoid predators or to hunt prey.
5. **Mimicry** - Animals imitate other animals to avoid predators.

Physiological Adaptation

1. **Torpor** - Animals reduce their metabolic rate to conserve energy.
2. **Blubber** - Thick layer of fat that provides insulation for animals in cold environments.
3. **Antifreeze Proteins** - Proteins that prevent ice crystals from forming in the body.

Structural Adaptation

1. **Webbed Feet** - Help animals swim in water.
2. **Spines** - Protect plants from herbivores.
3. **Sharp Claws** - Help animals climb trees or catch prey.

Physics & Chemistry Concepts

1. **Electrical Charges** -
 - **Positive Charge** - Protons.
 - **Negative Charge** - Electrons.
 - **Neutral Charge** - Neutrons.
2. **Types of Energy** - Kinetic, Potential, Chemical, Thermal, Electrical, Nuclear.
3. **Temperature Extremes of Gases** -
 - **Absolute Zero** - Lowest possible temperature.
 - **Boiling Point** - Temperature at which a liquid boils.
 - **Melting Point** - Temperature at which a solid melts.

Biomolecules & Nutrients

1. **Elements in Carbohydrates & Lipids** - Carbon, Hydrogen, Oxygen.
2. **Macromolecules (CP-N-L)**
 - **Carbohydrates** - Sugars, Starches, Cellulose. Provide energy.

- **Proteins** - Amino acids. Build and repair tissues.
 - **Nucleic Acids** - DNA, RNA. Carry genetic information.
 - **Lipids** - Fats, Oils, Waxes. Store energy, insulate.
3. **Functions of Proteins** - Build and repair tissues, enzymes, hormones.
 4. **Nucleic Acids** - Store and transmit genetic information.
 - **DNA** - Double helix. Carries genetic code.
 - **RNA** - Single strand. Involved in protein synthesis.

Micronutrients (Vitamins & Deficiencies)

1. **Vitamin A** - Vision.
2. **Vitamin B** - Metabolism.
3. **Vitamin C** - Immune system.
4. **Vitamin D** - Bone health.
5. **Vitamin E** - Antioxidant.
6. **Vitamin K** - Blood clotting.

Deficiency Diseases

1. **Scurvy** - Vitamin C deficiency.
2. **Rickets** - Vitamin D deficiency.
3. **Beriberi** - Vitamin B1 deficiency.
4. **Pellagra** - Niacin (Vitamin B3) deficiency.

Blood Types & Compatibility

Blood Types - A, B, AB, O.

- **O-Negative** – **Universal donor**; can be given to **anyone**, but can only receive from **O-**; critical in **trauma** and **emergencies**.
- **O-Positive** – **Most common** type; can donate to all **Rh-positive** types (**A+, B+, AB+, O+**).
- **A-Negative** – **Rare type** (only **6%**); can donate to **A-** and **AB-**, but can only receive **A-** or **O-**.
- **A-Positive** – **Second most common** type; can receive from **A** and **O** types with **positive Rh**.
- **B-Negative** – **Very rare**; can donate to **B-** and **AB-**; important for **rare blood banks**.
- **B-Positive** – Found in about **9%** of people; can give to **B+** and **AB+**, and receive from **B** or **O** types.
- **AB-Negative** – **Extremely rare (<1%)**; **universal plasma donor**, but needs **Rh-negative** blood only.
- **AB-Positive** – **Universal recipient**; can receive **any blood type safely**, but also the **rarest** of all.

Coal Ranks (L-B-A-G, from lowest to highest)

1. **Lignite** - Lowest rank, brown coal.
2. **Bituminous** - Most common, black coal.
3. **Anthracite** - Highest rank, hard coal.
4. **Graphite** - A form of carbon, lubricant.

Atomic Structure (AP-N-E)

1. **Atomic Number** - Number of protons.
2. **Mass Number** - Number of protons and neutrons.
3. **Number of Electrons** - Equal to number of protons in a neutral atom.
4. **Number of Neutrons** - Mass number minus atomic number.

Biological Processes

1. **Activation** - Summer sleep to survive hot, dry conditions.
2. **Hibernation** - Deep sleep to survive cold conditions.
3. **Dormancy** - Period of inactivity in plants.
4. **Estivation** - Prolonged inactivity in animals during hot periods.
5. **Homeostasis** - Maintaining stable internal environment.
6. **Decomposition** - Breaking down dead organisms.
7. **Carbon Removal** - Removing carbon from atmosphere.
8. **Nitrogen Fixation** - Converting atmospheric nitrogen into usable forms.
9. **Photosynthesis** - Plants convert light energy into chemical energy.
10. **Cellular Respiration** - Cells release energy from food.

Brainwaves (A-B-T-D)

1. **Alpha Waves** - Relaxation, meditation.
2. **Beta Waves** - Alert, active, focus.
3. **Theta Waves** - Sleep, deep relaxation.
4. **Delta Waves** - Deep sleep.

Forest Structures and Processes

- **Canopy** - Uppermost layer of forest.
- **Understory** - Layer beneath canopy.
- **Forest Floor** - Ground layer.
- **Decomposition** - Breaking down dead organic matter.
- **Nutrient Cycling** - Movement of nutrients through ecosystem.

Types of Ecosystems

- **Terrestrial** - Land-based ecosystems.
- **Aquatic** - Water-based ecosystems.

Relationships between Organisms

1. **Producers** - Organisms that produce their own food.
2. **Consumers** - Organisms that eat other organisms.
3. **Decomposers** - Organisms that break down dead organisms.

Energy Sources & Biological Role

1. **Sunlight** - Primary energy source for Earth.
2. **Photosynthesis** - Converts light energy to chemical energy.
3. **Chemosynthesis** - Organisms produce food without sunlight.

Parts of People & Biological Relevance

1. **Stomach** - Digests food.
2. **Small Intestine** - Absorbs nutrients.
3. **Large Intestine** - Absorbs water.
4. **Lungs** - Gas exchange.
5. **Heart** - Pumps blood.
6. **Brain** - Controls body functions.
7. **Kidneys** - Filter blood.
8. **Liver** - Detoxifies blood.
9. **Pancreas** - Produces insulin.

Key Science Concepts

1. **Biotic** - Living (e.g., animals, plants, fungi, bacteria).
2. **Abiotic** - Non-living (e.g., water, air, soil, sunlight, temperature).
3. **Microscopic** - Too small to be seen with the naked eye.
4. **Macroscopic** - Large enough to be seen with the naked eye.
5. **Endemic** - Native to a specific region (e.g., Tamaraw in Mindoro, Philippine Eagle in Mindanao).
6. **Indigenous** - Native to a specific region, not necessarily endemic (e.g., plants and animals that live in Arctic waters).
7. **Exotic** - Not native to a specific region, introduced from elsewhere.
8. **Invasive** - Introduced species that spread aggressively and cause harm.

Human Embryonic Development (S-Z-C-E-F)

1. **Sperm** - Male reproductive cell.
2. **Egg (Ovum)** - Female reproductive cell.
3. **Fertilization** - Fusion of sperm and egg, forms a zygote.
4. **Zygote** - Fertilized egg.
5. **Cleavage** - Rapid cell division of zygote.
6. **Blastulation** - Formation of a blastula (hollow ball of cells).
7. **Gastrulation** - Formation of germ layers.
8. **Embryo** - Early developmental stage of a multicellular organism.
9. **Fetus** - Later developmental stage, before birth.

Level of Biological Organization

1. **Cell** - Basic structural and functional unit of life.
2. **Tissue** - Group of similar cells working together.
3. **Organ** - Group of tissues working together.
4. **Organ System** - Group of organs working together.
5. **Organism** - A complete living being.
6. **Population** - Group of same species living in an area.
7. **Community** - Different populations living in an area.
8. **Ecosystem** - Community and its non-living environment.
9. **Biome** - Large region characterized by climate, vegetation, and wildlife.
10. **Biosphere** - All ecosystems on Earth.

Chemistry Concepts

- 1. **Carbohydrates** - Primary source of energy (e.g., sugars, starches).
- 2. **Proteins** - Build and repair tissues (e.g., enzymes, hormones).
- 3. **Lipids** - Store energy, insulate (e.g., fats, oils).
- 4. **Nucleic Acids** - Store genetic information (e.g., DNA, RNA).

Phases of Mitosis (PMAT)

- 1. **Prophase** - Chromosomes condense.
- 2. **Metaphase** - Chromosomes align at the center.
- 3. **Anaphase** - Sister chromatids separate.
- 4. **Telophase** - New nuclei form.

Structures in Insects

- 1. **Exoskeleton** - Outer covering, provides support and protection.
- 2. **Wings** - Used for flight.
- 3. **Legs** - Used for walking, jumping.
- 4. **Antennae** - Sensory organs.
- 5. **Compound Eyes** - Vision.
- 6. **Mandibles** - Jaws

Photosynthesis & Plant Respiration

- 1. **Photosynthesis** - Plants take in CO2 and release O2.
- 2. **Respiration** - Plants take in O2 and release CO2.

Blood

- 1. **Plasma** - Liquid part of blood.
- 2. **Red Blood Cells** - Transport oxygen.
- 3. **White Blood Cells** - Fight infection.
- 4. **Platelets** - Help blood clot.

Oceans & Marine Life

- 1. **Ocean Zones** - Zones based on light and depth.
- 2. **Coral Reefs** - Underwater ecosystems.
- 3. **Marine Mammals** - Mammals that live in the ocean.
- 4. **Fish** - Aquatic vertebrates.
- 5. **Invertebrates** - Animals without backbones.

Ecosystems

- 1. **Food Web** - Interconnected food chains.
- 2. **Food Chain** - Transfer of energy from one organism to another.
- 3. **Producers** - Make their own food.
- 4. **Consumers** - Eat other organisms.
- 5. **Decomposers** - Break down dead organisms.

Ecosystems

- 1. **Diffusion** - Movement of molecules from high to low concentration.
- 2. **Osmosis** - Movement of water across a semipermeable membrane.
- 3. **Active Transport** - Movement of molecules against concentration gradient.
- 4. **Facilitated Diffusion** - Movement of molecules with the help of protein channels.

Equinox & Solstice Table

Event	Northern Hemisphere	Southern Hemisphere
Spring Equinox	Mar. 20	Sep. 22
Summer Solstice	Jun. 21	Dec. 21
Autumn Equinox	Sep. 22	Mar. 20
Winter Solstice	Dec. 21	Jun. 21

Fungi

- **Yeasts** - Single-celled fungi.
- **Molds** - Multicellular fungi.
- **Mushrooms** - Fungi with fruiting bodies.

Pioneers in Science

- **Father of Computer** - Charles Babbage.
- **Father of Modern Computing** - Alan Turing.
- **Father of Artificial Intelligence** - John McCarthy.
- **Father of Robotics** - Joseph Engelberger.

Important Earth & Space Features

- 1. **Arctic** - North Pole region.
- 2. **Antarctic** - South Pole region.
- 3. **Equator** - Divides Earth into Northern and Southern Hemispheres.
- 4. **Prime Meridian** - Divides Earth into Eastern and Western Hemispheres.
- 5. **International Date Line** - Separates calendar days.

Abundant Elements in the Human Body (CHON)

- **Carbon**
- **Hydrogen**
- **Oxygen**
- **Nitrogen**

Limbic System - Emotional Brain

- 1. **Amygdala** - Emotions (e.g., fear, anger).
- 2. **Hippocampus** - Memory.
- 3. **Hypothalamus** - Regulates basic functions (e.g., hunger, thirst).
- 4. **Thalamus** - Relays sensory information.
- 5. **Cingulate Gyrus** - Involved in emotion and behavior.
- 6. **Basal Ganglia** - Involved in motor control.

Macromolecules (CPLN)

- 1. **Carbohydrates** - Energy source (glucose, starch).
- 2. **Proteins** - Building blocks of life (amino acids).
- 3. **Lipids** - Store energy (fats, oils).
- 4. **Nucleic Acids** - Genetic material (DNA, RNA).

Types of Doctors (CARP-GP-O-N-O-P-E-S)

- 1. **Cardiologist** - Specializes in heart and blood vessels.
- 2. **Pulmonologist** - Focuses on lung and respiratory system.
- 3. **Gastroenterologist** - Treats digestive system.
- 4. **Oncologist** - Specializes in cancer.
- 5. **Neurologist** - Treats brain, spinal cord, and nervous system disorders.
- 6. **Orthopedist** - Specializes in bones, joints, muscles, and ligaments.
- 7. **Pediatrician** - Specializes in child health.
- 8. **Ophthalmologist** - Specializes in eye diseases and eye surgery.
- 9. **Dermatologist** - Treats skin, hair, and nail disorders.
- 10. **Endocrinologist** - Treats hormone and endocrine system disorders.
- 11. **Psychiatrist** - Specializes in mental health.
- 12. **Surgeon** - Performs surgery.

SCIENCE, TECHNOLOGY, AND SOCIETY

History of Science & Technology in the World

Stone Age

- Tools from stone, wood, bone; used for hunting.
- Early humans: **Neanderthals** and **Denisovans**.

Three Main Period of Stone Age

- 1. **Paleolithic Period**
(Old Stone Age, 2.5 MYA – 10,000 B.C.)
 - Nomadic; lived in caves.
 - Basic stone tools, controlled fire, hunted and gathered.
- 2. **Mesolithic Period**
(Middle Stone Age, 10,000 B.C. – 8,000 B.C.)
 - Permanent settlements near rivers.
 - Fishing, agriculture, smaller stone tools.
- 3. **Neolithic Period**
(New Stone Age, 8,000 B.C. – 3,000 B.C.)
 - Advanced farming, polished tools, domestication of animals.
 - Home construction and art.

Bronze Age (3,000 B.C. – 1,300 B.C.)

- **Smelting**: blending copper and tin for stronger tools.
- Rise of states/kingdoms: **Sumer**, **Babylonia**, **Athens**.

Iron Age (1200 B.C. – 600 B.C.)

- **Hittites** created steel by heating iron with carbon.
- Mass production, armored Persian cavalry.

Antiquity (600 BCE – 529 CE)

- Rise of **Greek civilization**, **natural philosophers** emerged.

Pre-Socratic Thinkers

- 1. **Thales**: Earth floats on water.
- 2. **Anaximander**: Evolution of life.
- 3. **Anaximenes**: Air = primary substance.
- 4. **Heraclitus**: Fire is essence of change.
- 5. **Leucippus**: Idea of atom.
- 6. **Democritus**: Atoms are indivisible.
- 7. **Pythagoras**: Earth is spherical.
- 8. **Hippocrates**: Father of Medicine; Hippocratic Oath.

Socratic Philosophers

- 1. **Socrates**: Socratic Method.
- 2. **Plato**: Academy, used geometry over observation.
- 3. **Aristotle**: Lyceum, fifth element aether, Father of Biology, inductive method.

Other Notables

- 1. **Aristarchus**: Heliocentric idea.
- 2. **Ptolemy**: Geocentric model.
- 3. **Herophilus**: Father of Anatomy.
- 4. **Erasistratus**: Founder of Physiology.
- 5. **Euclid**: Father of Modern Geometry.
- 6. **Archimedes**: Father of Mathematics.

Middle Ages (476 – 1450 A.D.)

- **Early/Dark Middle Ages (476 – 1000 A.D.)**
 - ✓ Science slow due to Catholic Church.
 - ✓ **Islamic Empire** preserved ancient texts.
 - ✓ Discoveries: **Heavy plough**, **Horse collar**, **gunpowder**, **compass**, **decimal system**.
 - ✓ **Arab contributions**: borax, sal-ammoniac, Toledan tables.
- **High Middle Ages (1000 – 1450 A.D.)**
 - ✓ **St. Thomas Aquinas**: Scholasticism.
 - ✓ **Roger Bacon**: Experimentation.
 - ✓ **Water Wheel**: Key mechanical innovation.
 - ✓ **Black Death**: Yersinia pestis via fleas.

Modern Age (1453 – 1659 AD)

- **Renaissance**
 - ✓ **Francis Bacon**: Popularized Scientific Method.
 - ✓ **Gutenberg**: Printing Press.
 - ✓ **Paracelsus**: Laudanum (painkiller).
 - ✓ **Leonardo da Vinci**: Engineering, parachutes, concept of inertia.
 - ✓ **Tartaglia**: 45° canon angle, Pascal's triangle.
- **Scientific Revolution**
 - ✓ **Copernicus**: Heliocentric theory (De revolutionibus).
 - ✓ **Kepler**: Planetary motion laws.
 - ✓ **Galileo**: Telescope, motion laws, moon's craters, Jupiter's moons.

- ✓ **Tycho Brahe:** Geo-heliocentric model, supernova.
- ✓ **John Ray:** "Species" concept.
- ✓ **Linnaeus:** Binomial nomenclature, taxonomy.
- ✓ **Vesalius:** Human anatomy drawings.
- ✓ **Harvey:** Circulatory system.
- ✓ **Malpighi:** Capillaries.
- ✓ **Pascal:** Mechanical calculator.
- ✓ **Torricelli:** Barometer and vacuum.

• Enlightenment (1735–1819)

- ✓ **Isaac Newton:** Motion laws, gravity, calculus.
- ✓ **Cassini:** First modern map.
- ✓ **Watt:** Steam engine.
- ✓ **Lamarck:** Evolution theory.
- ✓ **Darwin:** Natural selection (Origin of Species).
- ✓ **Freud:** Psychoanalysis (id, ego, superego).
- ✓ **Alan Turing:** Turing machine, Turing Test.

S&T in the Philippines

Pre-Colonial

- Used metals, tools, medicinal plants.
- Astronomical prediction of seasons.
- Writing, measuring systems, boats (caracoa), and trade with China & Vietnam.

Spanish Period

- Formal education began (e.g., Colegio de San Ildefonso).
 - **UST:** center of medicine/pharmacy.
 - **San Juan de Dios Hospital** (oldest).
 - **Manila Observatory:** typhoon prediction.

American & Post-Commonwealth

- **UP system** for higher education.
- **Science Act of 1958, NSDB, Institute of Science, NRCP.**

American & Post-Commonwealth

- Science high schools, scholarships.
- Created: **PAG-ASA, IRRI, PNOC, Phil. Coconut Research Institute.**

Fifth Republic

- **Cory Aquino:** DOST, STMP (1991–2000), RA 6655.
- **Ramos:** STAND, RA 8439 (Magna Carta), R&D emphasis.
- **Estrada:** Clean Air Act (RA 8749), E-Commerce Act (RA 8792).
- **Arroyo:** Golden Age of S&T, Biofuels Act (RA 9367).
- **Aquino III:** Recognized scientists like Trono, Alcala, Barba, Del Mundo, Banatao.

STS and the Human Condition

Human Flourishing

- **Eudaimonia:** Peak happiness (Aristotle).
- Western = individual; Eastern = community.
- **Science as a method and social endeavor** (problem-solving, motivation, division of labor).

The Good Life

- **Materialism:** Atomists (Leucippus, Democritus).
- **Hedonism:** Epicurus (pleasure = goal).
- **Stoicism:** Virtue = happiness (Aurelius).
- **Theism:** Monotheism & polytheism.
- **Humanism:** Self-determination, stewardship.

Technology and Humanity

- **Inventions:** TV (Nipkow), mobile phones (Cooper), computer (Babbage).
- **Roles:** Communication, info, recreation, propaganda.
- **Robots:**
 - Types: AMRs, AGVs, Cobots, Humanoids.
 - Ethical issues: privacy, job loss, AI bias, robot rights, singularity.

Specific STS Issues

1. Information Age

- **Claude Shannon:** Father of Info Age.
- Eras: Pre-Mechanical, Mechanical, Electro-Mechanical, Electronic.
- Benefits: Globalization, new jobs, low cost, connectivity.

2. Gene Therapy

- Technique: Fix defective genes.
- **Ex vivo vs. In vivo.**
- Examples: CRISPR, CAR-T, Spinraza, Luxturna.
- Ethics: Consent, justice, embryo research.

3. Biodiversity

- Affects nutrition, disease.
- 6th Mass Extinction ongoing.
- Top 10 issues: Pollution, global warming, deforestation, etc.

4. Nanotechnology

- **Feynman** = Father of Nanotech.
- Applications: solar, body armor, food packaging.
- Uses: diagnostics, agriculture, construction, pollution control.

ADDITIONAL STS INFORMATION

- The term "**scientist**" was coined by **William Whewell** in 1833. Before that, they were called "natural philosophers."
- **Gregor Mendel's work on genetics** was ignored for 35 years until it was rediscovered around 1900.

- **Francis Bacon** is often called the “Father of the Scientific Method” — he emphasized experimentation over pure reason.
- The **Green Revolution** in the 1940s–1960s helped prevent global famines by introducing high-yield crops and fertilizers.
- **Alfred Wegener’s continental drift theory** was rejected in his lifetime — only accepted decades later as “plate tectonics.”
- The word **vaccine** comes from the Latin **vacca** (cow) because in 1796, English doctor **Edward Jenner** used *cowpox* pus to protect people from **smallpox**—an early form of vaccination already practiced in **China** and the **Ottoman Empire** before modern medicine.
- The **first programmable computer**, the Z3, was built in Nazi Germany by Konrad Zuse in 1941.
- The term “**technology**” comes from the Greek word *techne*, meaning craft or art — not just machines.
- In ancient Philippines, **baybayin** was a pre-colonial writing system that used syllabic characters.
- The **first human organ transplant** (kidney) was done in 1954 between identical twins.
- **Rachel Carson’s book "Silent Spring" (1962)** helped launch the global environmental movement.
- In Japan, the term “**karoshi**” means death from overwork — a tech-society issue.
- The **Space Race** was not just science — it was a Cold War political move to show power.
- **Science denialism** (e.g., anti-vaccine, flat Earth) is a sociopolitical problem, not just a lack of knowledge.
- **Galileo Galilei** was not actually imprisoned for saying the Earth moves(**heliocentric**) — he was placed under house arrest.
- The **first genetically modified crop** approved for consumption was the **Flavr Savr tomato** in 1994 (U.S.).
- In the Philippines, **RA 8439 (Magna Carta for Scientists)** aims to support Filipino scientists with better pay and rights.
- **Thomas Kuhn’s "The Structure of Scientific Revolutions"** introduced the idea of “paradigm shift” in how science changes.
- The **Philippine Rice Research Institute (PhilRice)** is a key player in making agriculture more tech-based locally.
- **Biotechnology** in the Philippines is regulated by the **National Committee on Biosafety of the Philippines (NCBP)**.
- **Google** was started as a university research project by two PhD students, Larry Page and Sergey Brin.
- **Nanotechnology** involves working with materials at the atomic level — 1 nanometer = 1 billionth of a meter.
- The term “**cyborg**” was first used in 1960 to describe a human-machine system for space travel.
- The **Philippine Science High School (PSHS)** system was established in 1964 to nurture young scientists.
- **CRISPR-Cas9** is a real-life gene-editing tool inspired by bacteria — parang “molecular scissors” for DNA.
- The term “**Artificial Intelligence (AI)**” was first used in 1956 at a conference in Dartmouth College.
- On **March 29, 1994**, **Benjamin Tan** established the Philippines' **first connection to the Internet** at a network center in Makati City.
- In the Philippines, **Project NOAH** (Nationwide Operational Assessment of Hazards) uses tech to warn against disasters.
- The **first space tourist** was **Dennis Tito**, who paid \$20 million in 2001 to go to space.
- **4th Industrial Revolution (4IR)** involves AI, robotics, IoT, and biotechnology merging into everyday life.
- **Plastic waste in oceans** partly comes from “sachet economies” — especially common in lower-income countries.
- **Bioethics** deals with issues like cloning, euthanasia, and GMOs — asking not just “Can we?” but “Should we?”
- **Fe del Mundo**, a Filipina pediatrician, was the first woman admitted to Harvard Medical School — and she invented a bamboo incubator for rural use.
- The **first Filipino satellite**, Diwata-1, was launched into space in 2016 under the PHL-Microsat Program.
- The term “**scientific revolution**” refers to the period (roughly 1543–1700) when figures like Galileo and Newton transformed how people understood nature.
- **Leonardo da Vinci**, known for art, also sketched prototypes of tanks, helicopters, and underwater suits — centuries before they existed.
- **Benjamin Franklin** invented the lightning rod, bifocal glasses, and the Franklin stove — he wasn’t a career scientist, but a curious polymath.

- The **first microscope** was invented around 1590 by Dutch eyeglass makers Hans and Zacharias Janssen.
- **Marie Curie** was the first person (and only woman) to win Nobel Prizes in two different sciences: Physics and Chemistry.
- **The first vaccine ever made** was for smallpox in 1796, created by Edward Jenner using cowpox.
- **The first telephone call** was made by Alexander Graham Bell in 1876 — he said, “Mr. Watson, come here, I want to see you.”
- **The first computer bug** was literally a bug — a moth found stuck in a computer relay in 1947.
- **Charles Darwin’s theory of evolution** wasn’t widely accepted at first — it clashed with religious beliefs in the 1800s.
- The **term “technology”** used to mean “art or craft” — it only became associated with machines in the 20th century.
- **Gregor Mendel**, the father of genetics, was an Austrian monk who studied peas in his garden.
- The **Philippine Institute of Volcanology and Seismology (PHIVOLCS)** was founded in 1982 to monitor earthquakes and volcanoes.
- **Ramon Barba**, a Filipino scientist, developed the mango flowering technology using potassium nitrate — boosting PH mango exports.
- **Arturo Alcaraz**, a volcanologist, helped pioneer geothermal energy use in the Philippines as early as the 1960s.
- **The term “eugenics”**, now discredited, was a science movement in the early 1900s promoting selective breeding of humans.
- **The telescope** was first used for astronomy by Galileo in 1609 — even though it was originally invented for ships.
- **Isaac Newton** invented calculus independently around the same time as Leibniz — leading to a fierce priority dispute.
- **The steam engine** kickstarted the Industrial Revolution in the 18th century by transforming factories and transport.
- **The scientific method** we know today was shaped by philosophers like Francis Bacon (induction) and René Descartes (deduction).
- **In the Philippines**, the first science high school was the Manila Science High School, founded in 1963.
- **The Nobel Prize** was founded by Alfred Nobel, a chemist who invented dynamite — he wanted to leave a positive legacy.
- **DNA’s double helix** structure was discovered in 1953 by Watson and Crick — but much of the data came from Rosalind Franklin.
- **The concept of zero** came from Indian mathematics, spreading to the Islamic world and then to Europe.
- **The compass, gunpowder, paper, and printing** are the Four Great Inventions of ancient China.
- **Johannes Gutenberg** revolutionized knowledge access by inventing the movable-type printing press in the 15th century.
- **The Philippines’ first weather bureau** was established in 1865 during the Spanish colonial period.
- **Fidel Ramos signed RA 8439 (1997)** or the Magna Carta for Scientists to give better benefits to Filipino S&T workers.
- **The first weather satellite**, TIROS-1, was launched by the US in 1960 — it could take pictures of cloud systems.
- **Wernher von Braun**, a Nazi rocket scientist, later became a leading figure in NASA’s Apollo program.
- **The ancient Filipinos** practiced metallurgy and boat-building — the balangay boats prove precolonial maritime technology.
- **The term “cloning”** became mainstream in 1996 after the birth of Dolly the sheep — the first cloned mammal.
- **Carbon dating**, used to measure age of fossils, was developed in the late 1940s by Willard Libby.
- **The first man-made object in space** was Sputnik 1, launched by the USSR in 1957 — triggering the Space Race.
- **The first AI chatbot** was **ELIZA**, made in 1966 — it mimicked a therapist using simple pattern matching.
- **The largest radio telescope** in the world today is China’s **Five-hundred-meter Aperture Spherical Telescope (FAST)** — used to search for extraterrestrial signals and cosmic phenomena.